

SYSTEM REQUIREMENTS SPECIFICATION DIL

Digital Interlocking OBB
RAMS

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Verified by	Rotter F. / QAM	
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LOG OF CHANGES			
Edition	Date	Author	Change Description
01 RL	2025-11-07	Hitachi Rail GTS Austria / EJT RM G. Eber	First edition. Details see 4.1.1

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1. INTRODUCTION

1.1 PURPOSE

The present document contains the system requirements of the Digital Interlocking (DIL). The system is developed in the context of the “Generic Application Digital Interlocking OBB”. This part of the SSS focuses on the **[RAMS]** requirements derived from [PH_PART_D2.2].

1.2 INTENDED AUDIENCE

- Product Design Authority
- Technical Project Managers
- Solution Engineering Managers
- Product Line Manager
- Test Manager
- Validator
- Verifier
- Safety Assurance Manager
- Security Assurance Manager
- Quality Assurance Manager
- RAM Assurance Manager
- System Architects

1.3 DISTRIBUTION

A new version of the document is distributed to all persons according to chapter 1.2

1.4 DOCUMENT MANAGEMENT

See Requirements Management Plan [DIL_RMP] for a description of the SSS document management.

1.5 REFERENCED DOCUMENTS

[SP_GLOSS]	SharePoint Project Glossary Abbreviations for Glossary without duplicates.xlsx
[EL_GLOSS]	Glossary: Definitions, Acronyms and Abbreviations - ELEKTRA 2.x System, 3BU 15000 0081 DCAPA, Ed. 06
[DIL_RMP]	Requirements Management Plan – Digital Interlocking OBB, 3BU 17300 1207 DUAPA, Ed. 01
[LH_PART_D]	DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte, 3BU 17300 0004 DSAPC, Ed. 04

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[PH_PART_D2.2]	Pflichtenheft Digital Interlocking OBB - D2.2 Teilpflichtenheft RAMS, 3BU 17300 0022 DTAPC, Ed. 02
[PH_PART_D2.3]	Pflichtenheft Digital Interlocking OBB - D2.3 Teilpflichtenheft Allgemeine Anforderungen, 3BU 17300 0023 DTAPC, Ed. 02
[PH_PART_D2.5]	Pflichtenheft Digital Interlocking OBB - D2.5 Teilpflichtenheft Teilsystem DSTW-ZE, 3BU 17300 0025 DTAPC, Ed. 02
[PH_PART_D2.6]	Pflichtenheft Digital Interlocking OBB - D2.6 Teilpflichtenheft Teilsystem Weiche, 3BU 17300 0026 DTAPC, Ed. 02
[CMP]	Configuration Management Plan, 3BU 15000 3338 QMAPA, Ed.11
[TECHNICAL_CONCEPTS]	Digitales Stellwerk - Technische Konzepte, 3BU 17300 0010 DSAPC, Ed.04
[ARCHITECTURE_DRAWING]	Top Level Architecture Drawing - Generic Application - Digital Interlocking OBB, 3BU 17300 1218 DSAPA, Ed.01
[BFZ_AUSFALLKONZEPT]	Anforderungsspezifikation BFZ-Ausfallszenarien Leit- und Sicherungstechnik, 3BU 51500 2003 DSAPC, Ed.01
[RAM]	Project specific RAM Plan - Generic Application Digital Interlocking OBB - Solution Element VIL 3BU_17300_1209_DUAPA, Ed.01
[R10]	Eisenbahnanlagen Schwerpunktkonzept aus Sicht des Arbeitnehmer/innenschutzes
[RW 13.01.01]	Regelwerk Eisenbahnsicherungsanlagen
[RW 13.01.02]	Regelwerk Punktförmige Zugebeeinflussung
[RW 30.02]	Regelwerk Signalbuch
[RW 47.07.10]	Regelwerk Eisenbahnanlagen
[SSS_DIL_GENERAL]	System Requirements Specification DIL – General Requirements, 3BU 17300 1251 DTAPA Ed.01
[SSS_DIL_DATA_CENTER]	System Requirements Specification DIL - Data Center, 3BU 17300 126x DTAPA Ed.01 (Planned)
[SSS_DIL_SECURITY]	System Requirements Specification DIL - Security, 3BU 17300 1252 DTAPA Ed.01 (Planned)
[SSS_DIL_CU_1]	System Requirements Specification DIL – Central Unit – Part 1, 3BU 17300 1255 DTAPA Ed.01
[SSS_DIL_CU_2]	System Requirements Specification DIL – Central Unit – Part 2, 3BU 17300 1258 DTAPA Ed.01 (Planned)
[SSS_DIL_CU_3]	System Requirements Specification DIL – Central Unit – Part 3, 3BU 17300 1259 DTAPA Ed.01 (Planned)
[Low_Voltage]	Low Voltage Directive 2014/35/EU
[EMC]	EMC Directive 2014/30/EU
[RoHS 2]	Directive on hazardous substances 2011/65/EU (RoHS 2)
[Ecodesign]	Ecodesign Directive 2019/424/EU (Lot 9)
[EN 1991-1-4]	1991-1-4:2023 Eurocode 1: Einwirkungen auf Tragwerke Teil 1-4: Allgemeine Einwirkungen - Windlasten Nationale Festlegungen zu ÖNORM EN 1991-1-4 und nationale Ergänzungen 2023-04-01
[EN 50121-1]	EN 50121-1:2017 Bahnanwendungen - Elektromagnetische Verträglichkeit -- Teil 1: Allgemeines 2017-12
[EN 50121-4]	EN 50121-4:2016 +A1:2019 Railway applications – Electromagnetic compatibility - Part 4: Emission and immunity of the signalling and telecommunications apparatus

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[EN 50122]	EN 50122-1:2011 +A1:2011 +AC:2012 +A2:2016 +A3:2016 +A4:2017 Railway applications - Fixed installations - Electrical safety, earthing and the return circuit
[EN 50124-2]	EN 50124-1: 2017 Railway applications - Insulation coordination – Part 2 Overvoltage and related protection
[EN 50125-3]	EN 50125-3:2003 +AC:2010: Railway applications - Environmental conditions for equipment - Part 3: Equipment for signalling and telecommunications - Part 1: Protective provisions against electrical shock
[EN 50126]	EN 50126-1:2017 – Railway Applications – The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS)
[EN 50129]	EN 50129:2018 +AC:2019 – Railway applications – Communication, signalling and processing systems – Safety related electronic systems for signalling
[EN 50159]	EN 50159:2010 +A1:2020 – Railway application – Communication, signalling and processing: Safety-related communication in transmission systems
[EN 50310]	EN 50310:2016+A1:2020 - Telecommunications bonding networks for buildings and other structures - 2020-05-03
[EN 50716]	EN 50716:2024 - Railway Applications – Requirements for software development - 2024-10-01
[EN 60529]	IEC 60529:1989/A2:2013/COR1:2019 EN 60529:1991/A2:2013/AC:2019-02 Degrees of protection provided by enclosures (IP Code) (IEC 60529:1989/A2:2013/COR1:2019) (Corrigendum) - 2019-07
[EN 61010-1]	IEC 61010-1:2010+COR:2011+A1:2016 (MOD)+A1:2016/COR1:2019 EN 61010-1:2010 6+A1:2019+A1:2019/AC:2019 Safety requirements for electrical equipment for measurement, control, and laboratory use Part 1: General requirements 2020-04
[EN 60721-3-4]	EN IEC 60721-3-4:2019 Classification of environmental conditions Part 3-4: Classification of groups of environmental parameters and their severities – Stationary use at non-weatherprotected locations (IEC 60721-3-4:2019) 2020-06
[EN 62262]	IEC 62262:2002+AMD1:2021 Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code) consolidated version 2021-09-03
[EN 62305]	IEC 62305-1:2024 Protection against lightning – Part 1: General principles 2024-09-12 IEC 62305-2:2024 Protection against lightning – Part 2: Risk management 2024-09-12 IEC 62305-3:2024 Protection against lightning – Part 3: Physical damage to structures and life hazard 2024-09-12 IEC 62305-4:2024 Protection against lightning – Part 4: Electrical and electronic systems within structures 2024-09-12
[EN 62368-1]	EN IEC 62368-1:2025 Audio/video, information and communication technology equipment Part 1: Safety requirements 2025-02-01
[Eu.Doc.20]	EULYNX Generic interface and subsystem requirements, CENELEC Phase: 4, Version: 4.0 (6.A), Approval date: 29.05.2024
[Eu.Doc.7]	EULYNX System definition, CENELEC Phase: 2, Version: 4.2 (2.A), Approval date: 29.05.2024

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[Eu.Doc.33]	[Eu.Doc.33], Interface specification SCI-LS, CENELEC Phase: 5, Version: 4.3 (0.A), Approval date: 29.05.2024
[Eu.Doc.37]	[Eu.Doc.37], Signal aspect table, CENELEC Phase: 4-5, Version: 2.2 (1.G), Approval date: 29.05.2024
[Eu.Doc.38]	[Eu.Doc.38], Interface specification SCI-P, CENELEC Phase: 5, Version: 4.2 (1.A), Approval date: 29.05.2024
[Eu.Doc.41]	[Eu.Doc.41], Requirements specification for SCI-ILS, CENELEC Phase: 4, Version: 4.3 (0.A), Approval date: 29.05.2024
[Eu.Doc.42]	[Eu.Doc.42], Interface specification SCI-ILS, CENELEC Phase: 5, Version: 4.3 (0.A), Approval date: 29.05.2024
[Eu.Doc.46]	[Eu.Doc.46], Interface specification SCI-IO, CENELEC Phase: 5, Version: 4.0 (3.A), Approval date: 29.05.2024
[Eu.Doc.47]	[Eu.Doc.47], Requirements specification for SCI-RBC, Cenelec Phase: 4, Version: 4.0 (3.A), Approval date:
[Eu.Doc.48]	[Eu.Doc.48], Interface specification SCI-RBC, Cenelec Phase: 5, Version: 4.0 (1.A), Approval date: 15.06.2023
[Eu.Doc.49]	[Eu.Doc.49], Requirements specification for SCI-CC, CENELEC Phase: 4, Version: 4.2 (1.A), Approval date:
[Eu.Doc.50]	[Eu.Doc.50], Interface specification for SCI-CC, CENELEC Phase: 5, Version: 4.3 (0.A), Approval date: 29.05.2024
[Eu.Doc.76]	[Eu.Doc.76], Interface definition and specification SMI, CENELEC Phase: 5, Version: 2.2 (0.A), Approval date: 29.05.2024
[Eu.Doc.94]	[Eu.Doc.94], Interface specification SDI Generic, CENELEC Phase: 5, Version: 4.2 (0.A), Approval date: 29.05.2024
[Eu.SAT.244]	Speed indication (10 km/h) to
[Eu.SAT.246]	Speed indication (150 km/h)
[Eu.SAT.247]	Speed indication dark - Commanded to dark
[Eu.SAT.346]	Intended dark - Commanded to dark
[Eu.SAT.351]	Not equipped - Signal is not equipped with an extension of the basic signal aspect
[Eu.SAT.352]	Not equipped - Signal is not equipped with a speed indicator
[Eu.SAT.355]	Speed announcement (10 km/h) to
[Eu.SAT.356]	Speed announcement (150 km/h)
[Eu.SAT.357]	Speed announcement dark - Commanded to dark
[Eu.SAT.358]	Not equipped - Signal is not equipped with a speed announcement
[Eu.SAT.359]	Direction announcements
[ASchG]	ArbeitnehmerInnenschutzgesetz
[EisbAV]	Eisenbahn-ArbeitnehmerInnenschutzverordnung
[AVO Verkehr]	Arbeitnehmerschutzverordnung Verkehr
[AM-VO]	Arbeitsmittelverordnung
[KennV]	Kennzeichnungsverordnung

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[Pflichtenheft EBO - Einheitliche Bedienoberfläche]	Einheitliche Bedienoberfläche, EBO2, Bedienkatalog 3BU 15000 4401 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Meldungskatalog 3BU 15000 4402 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Symbolkatalog Teil1 3BU 15000 4403 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Symbolkatalog Teil2 3BU 15000 4404 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Symbolkatalog Teil3 3BU 15000 4405 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Symbolkatalog Teil4 3BU 15000 4406 DTAPC Ed.03 Einheitliche Bedienoberfläche, EBO2, Typenkatalog 3BU 15000 4407 DTAPC Ed.02
RAHMENAUSSCHREIBUNG - ETCS STRECKENAUSRÜSTUNG - D.1 Lastenheft]	DSTW Streckenausrüstung OBB - Teil D Leistungsbeschreibung und Lastenhefte 3BU 17300 0004 DSAPC Ed. 04
[Schnittstellenbeschreibung X25oIP]	X25-over-IP RaSTA 3BU 15000 2377 DSAPC Ed. 02
[PH X25-IF]	Specification X25 Interface, 3BU 15000 4167 DTAPQ Ed.58
[PH X25 – BFZ Schnittstelle ÖBB]	X25-BFZ – Schnittstelle 3BU 15000 4974 DSAPC Ed.04
[LH – EBO2 / X25]	Lastenheft EBO2 / X25 1.0 01.04.2022

Table 1: Reference Documents

1.6 DEFINITIONS, ACRONYMS AND ABBREVIATIONS

See Definitions, Acronyms and Abbreviations [SP_GLOSS] and [EL_GLOSS].

Additional abbreviations not mentioned there: **see table below.**

TBD	TBD
-----	-----

Table 2: Abbreviations and Acronyms

1.7 STRUCTURE OF THE REQUIREMENTS

See [DIL_RMP] for a description of the SSS requirement structure, the used attributes and their meaning.

1.8 INFORMATION FOR DOCUMENT GENERATION

The present document was generated by using IBM™ Rational DOORS with following settings:

DOORS/RMF Version:	DOORS 9.7.1.0/RMF 6.1.0.9
DOORS/RMF Module Name:	EL4 SSS
DOORS/RMF Module Version (Baseline):	1.0
DOORS/RMF Module Path:	/ELEKTRA 4.0/Specification
DOORS/RMF Export View:	WEXP SSS 1257 (DIL)

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DOORS/RMF Current Filter:	(TA Req Status != Obsolete) AND (TA SRS includes 1257)
DOORS/RMF Export Options 1:	2,false,0,0,false,false,0,true,false,
DOORS/RMF Export Options 2:	true,false,true,2,false,false,true,PostprocessAllStyles,\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_1257.log,
DOORS/RMF Export Options 3:	\\Thales-AT-Data\Thales-AT\PROJECTS\ReqMng\DOORS\Word Export\DOORS-EXPORT-TEMPLATES\EL4\SRS\Exports\WEXP_1257.log,true,true,false,false,false,0,true,4,
DOORS/RMF Export Options 4:	3,###,===,true,false,false,false,false,
Documentation-Vob Template Name:	3BU_17300_1257_DTAPA-Template

Table 3: Document Generation Settings

1.9 OPEN POINTS

Update of all figures in accordance with the documents [TECHNICAL_CONCEPTS] and [ARCHITECTURE_DRAWING].

Topic “Interface to DDSP” -> remains as open point for the first DIL baseline (phase 1).

Topic “Geo-Redundancy” -> remains as open point for the first DIL baseline (phase 1).

Topics that require further clarification with the customer.

Review values concerning “Allocation” with architect/design team.

Review values concerning “Verification” with test manager.

Check completeness of attribute “Origin”.

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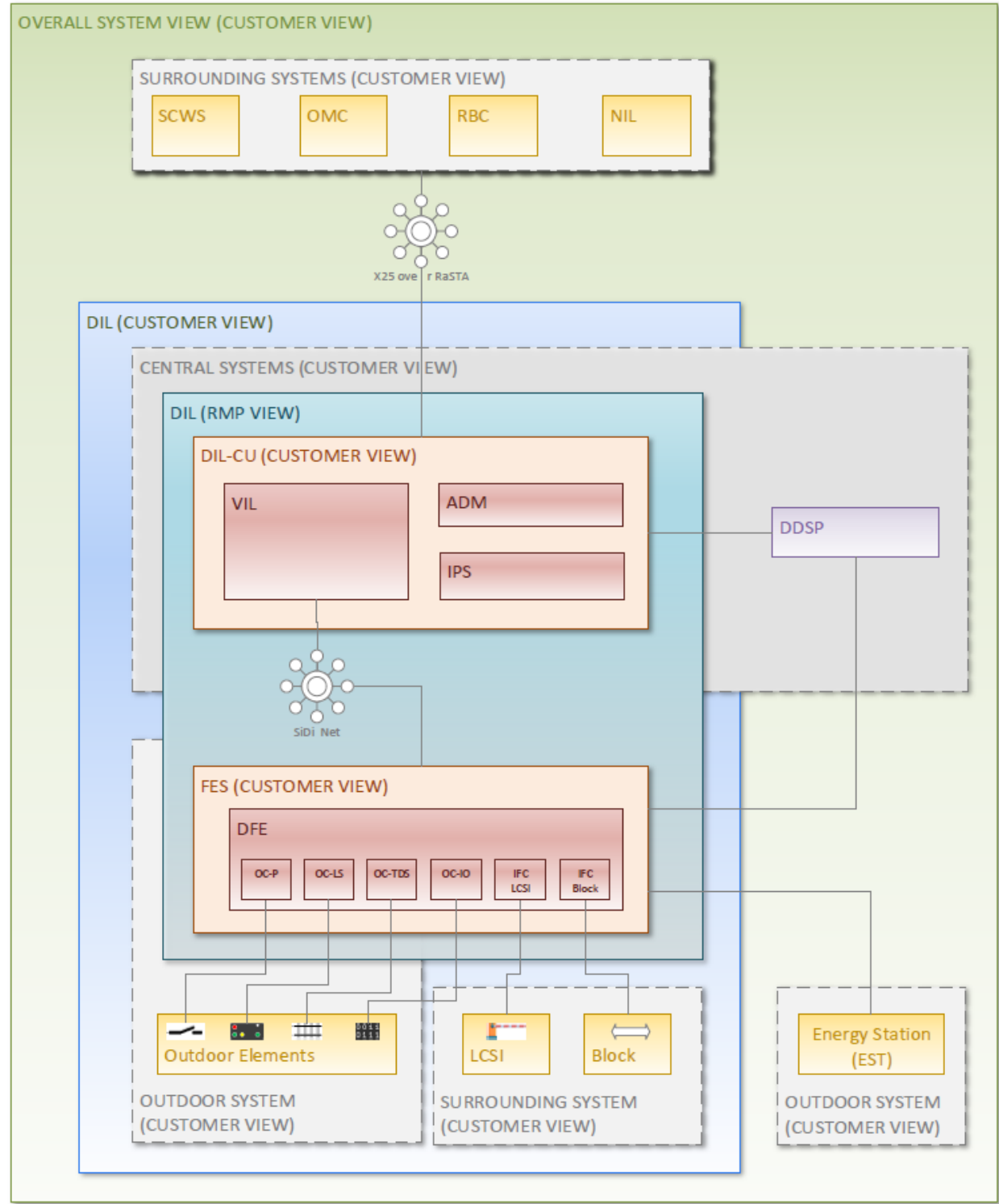
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2. OVERVIEW

The following picture gives an overview of the overall system and the [DIL] system considering both the technical project view and the customer’s view.



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Figure 1: System Overview

For an overview and a description of the system please refer to the document [TECHNICAL_CONCEPTS], chapter 1.

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3. REQUIREMENTS

3.1 D2.2 RAMS

3.1.1 GENERAL

3.1.1.1 ASSUMPTIONS

Assumption 1:

No safety level is required for power supply and maintenance interfaces.

Assumption 2:

No safety level is required for the [DDSP].

Assumption 3:

No safety level is required for all security functions in the sense of safety.

Assumption 4:

Concerning the availability calculation for the manual repair of a [fault] an [MRT] of 0,5h is assumed.

Assumption 5:

Concerning the availability calculation for the manual repair of an [error] an [MRT] of 1h is assumed.

3.1.1.2 MINIMUM USEFUL LIFE

The minimum useful life is specified in the service agreement and can be guaranteed by measures taken by the supplier, e.g. provision of spare parts/replacement products.

EL4-DIL-SSS-REQ-00299		Minimum Useful Life - Indoor System - COTS Components		(Accepted)
The minimum useful life of the (COTS) components of the interior system [DIL-CU] shall be at least 7 years (24 hours, 7 days a week).				
Comments:				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_16]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, IPS	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	2.1.0-1 Die Mindestnutzungsdauer der COTS-Komponenten der Innenanlage muss mindestens 7 Jahre betragen. Der Betrieb muss täglich ohne Betriebspause stattfinden.			
PH Text:	Eine sorgfältige Auswahl von qualitativ hochwertigen COTS-Komponenten gewährleistet einen langlebigen Betrieb der Innenanlage. Die Produktstrategie garantiert den ordnungsgemäßen Austausch und die Verfügbarkeit dieser HW-Komponenten mit der erforderlichen Funktionalität für den erforderlichen Zeitraum, so dass die Mindestnutzungsdauer von 7 Jahren gewährleistet ist.			

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EL4-DIL-SSS-REQ-00300 Minimum Useful Life - Continuous Utilisation		(Accepted)
The supplier shall guarantee the continuous utilisation of the [DIL] system within the minimum useful life according to [EL4-DIL-SSS-REQ-00299].		
<i>Comments:</i>		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_17]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, VIL, IPS, DFE	Verification: < will be added in a later document edition > See also: ==> [EL4-DIL-SSS-REQ-00299] Minimum Useful Life - Indoor System - COTS Components
<i>Issues/Decisions:</i>		
LH Text:	2.1.0-2 Die durchgehende Nutzung des Systems DSTW innerhalb der Mindestnutzungsdauer muss durch den AN gewährleistet werden.	
PH Text:	Eine sorgfältige Auswahl von qualitativ hochwertigen DSTW-Komponenten gewährleistet einen langlebigen Betrieb der Außenanlage. Die Produktstrategie garantiert den ordnungsgemäßen Austausch und die Verfügbarkeit dieser HW-Komponenten mit der erforderlichen Funktionalität für den erforderlichen Zeitraum, so dass die Betriebslebensdauer von 25 Jahren gewährleistet ist.	

EL4-DIL-SSS-REQ-01011 Minimum Useful Life - Outdoor System Components		(Accepted)
The minimum useful life of the components of the outdoor system shall be at least 25 years (24 hours, 7 days a week).		
<i>Comments:</i>		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_21]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, DFE	Verification: < will be added in a later document edition > See also: ==> [EL4-DIL-SSS-REQ-00343] Maintainability and Maintenance - Labelling - Serial Number
<i>Issues/Decisions:</i>		
LH Text:	2.3.0-1 Der AN der DSTW-Komponenten muss die Versorgung mit sämtlichen Ersatzteilen zur Aufrechterhaltung des Betriebes und der vollständigen Funktionalität für die Mindestnutzungsdauer von 7 Jahren der Komponenten der Innenanlage bzw. von 25 Jahren der Komponenten der Außenanlage sicherstellen.	
PH Text:	Der AN wird die Versorgung sämtlicher Ersatzteile zur Aufrechterhaltung des Betriebes und der vollständigen Funktionalität für die Mindestnutzungsdauer von 7 Jahren der Komponenten der Innenanlage bzw. von 25 Jahren der der Komponenten der Außenanlage sicherstellen.	

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EL4-DIL-SSS-REQ-00301 Minimum Useful Life - Extension		(Accepted)
The minimum useful life of the [DIL] components of the outdoor system shall be extendable once by 5 years.		
<i>Comments:</i>		
<i>Category:</i> DPR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_19] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i> ==> [EL4-DIL-SSS-REQ-00343] Maintainability and Maintenance - Labelling - Serial Number ==> [EL4-DIL-SSS-REQ-01011] Minimum Useful Life - Outdoor System Components	
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	2.2.0-1 Die Mindestnutzungsdauer der DSTW-Komponenten der Außenanlage kann einmalig um 5 Jahre verlängert werden.	
<i>PH Text:</i>	Über ein vom AN eingerichtetes FRACAS-System wird die Zuverlässigkeit der DSTW-Komponenten der Außenanlage gründlich untersucht und eine Verlängerung der Mindestnutzungsdauer um einmalig 5 Jahre ermöglicht.	

3.1.1.3 SPARE PARTS

EL4-DIL-SSS-REQ-00302 Supply for Spare Parts		(Accepted)
The supplier of the [DIL] components shall ensure the supply of all spare parts to maintain operation and full functionality for the minimum utilization period of the components of the indoor and outdoor system according to [EL4-DIL-SSS-REQ-00299] and [EL4-DIL-SSS-REQ-01011].		
<i>Comments:</i>		
<i>Category:</i> DPR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, IPS, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_21] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i> ==> [EL4-DIL-SSS-REQ-00299] Minimum Useful Life - Indoor System - COTS Components ==> [EL4-DIL-SSS-REQ-01011] Minimum Useful Life - Outdoor System Components	
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	2.3.0-1 Der AN der DSTW-Komponenten muss die Versorgung mit sämtlichen Ersatzteilen zur Aufrechterhaltung des Betriebes und der vollständigen Funktionalität für die Mindestnutzungsdauer von 7 Jahren der Komponenten der Innenanlage bzw. von 25 Jahren der Komponenten der Außenanlage sicherstellen.	
<i>PH Text:</i>	Der AN wird die Versorgung sämtlicher Ersatzteile zur Aufrechterhaltung des Betriebes und der vollständigen Funktionalität für die Mindestnutzungsdauer von 7 Jahren der Komponenten der Innenanlage bzw. von 25 Jahren der der Komponenten der Außenanlage sicherstellen.	

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3.1.1.4 OEVE/OENORM AND EULYNX

EL4-DIL-SSS-REQ-00303		Fulfillment of Standard EN 50126		(Accepted)
The requirements according to [OEVE/OENORM EN 50126] shall be fulfilled.				
Comments:		Compliance with the requirements of [OEVE/OENORM EN 50126] is documented in the RAM Plan and in the Safety Plan.		
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_23]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	2.4.0-1 Die Anforderungen gemäß [OEVE/OENORM EN 50126] müssen erfüllt werden.			
PH Text:	Entwicklung und Herstellung folgen dem RAMS-Managementprozess gemäß EN 50126. Details zur Erfüllung der Anforderungen aus [OEVE/OENORM EN 50126] werden in einem RAM Plan und einem Safety Plan dargelegt.			

EL4-DIL-SSS-REQ-00304		Fulfillment of Standard EN 50716		(Accepted)
The requirements according to [OEVE/OENORM EN 50716] shall be fulfilled.				
Comments: Note: [OEVE/OENORM EN 50716] was published on 1 October 2024, stipulating that [OEVE/OENORM EN 50128] will be withdrawn at the end of October 2026. Compliance with the requirements of [OEVE/OENORM EN 50716] is documented in the Safety Plan.				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_25]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	2.5.0-1 Die Anforderungen gemäß [OEVE/OENORM EN 50128] müssen erfüllt werden.			
PH Text:	Sicherheitsrelevante Software wird gemäß [OEVE/OENORM EN 50716] entwickelt. Details zur Erfüllung der Anforderungen aus [OEVE/OENORM EN 50716] werden in einem Safety Plan dargelegt.			

EL4-DIL-SSS-REQ-00305		Fulfillment of Standard EN 50129		(Accepted)
The requirements according to [OEVE/OENORM EN 50129] shall be fulfilled.				
Comments:		Compliance with the requirements of [OEVE/OENORM EN 50129] is documented in the Safety Plan.		
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_27]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, VIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	2.6.0-1 Die Anforderungen gemäß [OEVE/OENORM EN 50129] müssen erfüllt werden.			
PH Text:	Sicherheitsrelevante elektronische Systeme werden gemäß [OEVE/OENORM EN 50129] entwickelt. Details zur Erfüllung der Anforderungen aus [OEVE/OENORM EN 50129] werden in einem Safety Plan dargelegt.			

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EL4-DIL-SSS-REQ-00306		Fulfillment of EULYNX Standard Eu.Doc.16		(Accepted)
The requirements according to [EULYNX Specification] - (Eu.Doc.16) shall be fulfilled.				
Comments:	Note: For the fulfillment of the requirements specified in [Eu.Doc.93] see the [SCI] specific requirements.			
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_29]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, VIL, IPS, ADM, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	2.7.0-1 Die Anforderungen gemäß Eu.Doc.93 und Eu.Doc.16 müssen erfüllt werden.			
PH Text:	Die Anforderungen gemäß Eu.Doc.93 und Eu.Doc.16 werden erfüllt			

EL4-DIL-SSS-REQ-01478		Fulfillment of EULYNX Standard Eu.Doc.93		(Accepted)
The requirements according to [EULYNX Specification] - (Eu.Doc.93) shall be fulfilled.				
Comments:				
Category:	GCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_29]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, VIL, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	2.7.0-1 Die Anforderungen gemäß Eu.Doc.93 und Eu.Doc.16 müssen erfüllt werden.			
PH Text:	Die Anforderungen gemäß Eu.Doc.93 und Eu.Doc.16 werden erfüllt.			

3.1.2 RELIABILITY

3.1.2.1 DEFINITIONS

Reliability:

Ability to function as required without [failure] under given conditions and for a given time interval (definition according to [OEVE/OENORM EN 50126]).

[Fault]:

Refers to an incident that directly restrict system or operational performance and shall be resolved quickly.

[Error]:

Considered less critical for immediate operations, no operational impact.

[Failure]:

The state of an item in which it is unable to perform one or more of its required functions.

[Defect] classes:

→ **FK1**

Class 1: 'EMERGENCY'

[Fault]s that significantly hinder operations.

Example: Driving on sight/order, the DSTW being non-functional, or the failure of a central field element

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→ **FK2**

Class 2: 'CRITICAL'

[Fault]s that have a significant impact on parts (functional/local) of the operation or serious [error]s.

Example: Driving on sight/order in a subsection of the DSTW, loss of redundancy that risks total failure, or failure of a field element affecting a partial area

→ **FK3**

Class 3 'MAJOR'

Other [fault]s or significant [error]s.

Example: Only **minor impact** on operation, or a serious error that has **no impact** on operation, or the failure of individual Virtual Interlocking (VSTW) functions.

→ **FK4**

Class 4 'MINOR'

Other [error]s in the [DIL-CU] or outdoor system.

Example: **No operational impact**, an impact accepted by the Customer (AG), or failure of individual, less operationally relevant elements

3.1.2.2 IMPACT OF ERRORS AND FAULTS

EL4-DIL-SSS-REQ-00316		Reliability - Safe Design		(Accepted)			
The [DIL] system with its interfaces shall be designed in such a way that neither single [fault]s nor single [error]s lead to unsafe conditions.							
Comments:							
Category:		SAR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_35]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL, DFE		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		3.2.0-1 Das Gesamtsystem DSTW mit seinen Schnittstellen muss derart gestaltet werden, dass weder Fehler noch Störungen zu unsicheren Zuständen führen können.					
PH Text:		Das Gesamtsystem DSTW wird nach den sicherungstechnischen Grundsätzen eines Fail-Safe Systems entwickelt. Dabei kann weder ein einzelner Fehler noch eine einzelne Störung zu einem unsicheren Zustand führen.					

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EL4-DIL-SSS-REQ-00317 Reliability - Evidence by Calculation		(Accepted)
The supplier shall provide an evidence by calculation of the required reliability in accordance with [CENELEC] specifications.		
Comments: A mathematical proof of the required reliability will be provided to the customer in a RAM report in accordance with the specifications of a RAM plan and the requirements of the customer specifications.		
Category: DOR Originator: Customer Target Release: DIL_OBB_1.0.x DCA_OBB_3.1.0 Allocation: DIL, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_37] Scope: System Verification: < will be added in a later document edition > See also:	
Issues/Decisions:		
LH Text: 3.2.0-3 Ein rechnerischer Nachweis für die geforderte Zuverlässigkeit muss vom AN erbracht werden. Dieser muss gemäß CENELEC-Vorgaben erfolgen.		
PH Text: Ein rechnerischer Nachweis für die geforderte Zuverlässigkeit wird nach Vorgaben eines RAM Plans und der Lastenheftanforderungen in einem RAM-Report dem AG zur Verfügung gestellt werden.		

EL4-DIL-SSS-REQ-00319 Reliability - Error Reporting - OMC		(In negotiation)
[Fault]s, [error]s and [failure]s of the [DIL] system shall be reported to the [OMC] system for data recording via interface [IF-61] ([X25] over [IP] with [RaSTA]).		
Comments: == OPEN POINT == • Allocation of the faults, error und faults to the defect classes. • Clarification necessary, which components report to [OMC]		
Category: MDR Originator: Customer Target Release: DIL_OBB_1.0.x Allocation: DIL, VIL, IPS, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_41] Scope: System Verification: < will be added in a later document edition > See also:	
Issues/Decisions:		
LH Text: 3.4.0-1 Störungen, Ausfälle und Fehler des Systems DSTW müssen zur Datenregistrierung an das Umsystem BFZ gemeldet werden.		
PH Text: Störungen, Ausfälle und Fehler des Systems DSTW werden an die BFZ via IF-61 X25 over IP with RaSTA übertragen. Die Klassifizierung von Ereignissen gemäß dieser drei Kategorien ist noch zwischen AG und AN abzustimmen.		

EL4-DIL-SSS-REQ-00321 Reliability - Error Reporting - DDSP		(Accepted)
[Fault]s, [error]s and [failure]s of the [DIL] system shall be reported to the [DDSP].		
Comments:		
Category: MDR Originator: Customer Target Release: DIL_OBB_1.0.x DCA_OBB_3.1.0 Allocation: DIL, IPS, ADM, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_42] Scope: System Verification: < will be added in a later document edition > See also:	
Issues/Decisions:		
LH Text: 3.4.0-2 Störungen, Ausfälle und Fehler des Systems DSTW und seiner Schnittstellen müssen an das DDSP gemeldet werden.		
PH Text: Störungen, Ausfälle und Fehler der Innenanlage werden über die ADM an die DDSP gemeldet. Meldungen der Object Controller werden über die EULYNX Schnittstellen SDI und SSI an die DDSP übertragen.		

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EL4-DIL-SSS-REQ-00649 Reliability - Resolution of Faults - Responsibility (Accepted)			
[Fault]s shall not require the operator to take any maintenance-related substitute measures or maintenance activities.			
Comments: [Fault]s shall be resolved by the maintenance staff.			
Category:	DPR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_44]
Target Release:	DIL_OBB_1.0.x	Scope:	System
Allocation:	DIL	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	3.5.0-1 Fehler dürfen keine Instandhaltungstechnischen Ersatzmaßnahmen des Fdl erfordern und er darf in keine Instandhaltungstätigkeit einbezogen werden. Diese müssen durch die Instandhaltung behoben werden.		
PH Text:	Der AN sieht diese Anforderung als eine Begriffsdefinition für "Fehler". Ein Fehler führt zu keinen notwendigen Ersatzmaßnahmen durch den Fdl. Die Instandhaltung wird durch den AG durchgeführt.		

EL4-DIL-SSS-REQ-00322 Reliability - Replacement of HW Component - No operational Impact (Accepted)			
Replacement of failed/defective components shall have no adverse effect on the operation and reliability of the DIL system as a whole.			
Comments:			
Category:	DPR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_46]
Target Release:	DIL_OBB_1.0.x	Scope:	System
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	3.6.0-1 Ein Tausch ausgefallener/defekter Komponenten muss rückwirkungsfrei auf den Betrieb und die Zuverlässigkeit gegenüber dem Gesamtsystem DSTW sein.		
PH Text:	Der Tausch einer ausgefallenen HW-Komponente (korrektive Wartungsmaßnahme) wird rückwirkungsfrei auf den Betrieb durchgeführt unter Einhaltung der Zuverlässigkeit gegenüber dem Gesamtsystem DSTW.		

EL4-DIL-SSS-REQ-00990 Reliability - Single HW Failure - Diagnosis (Accepted)			
The [DIL] shall be able to report a single [failure] of a HW component to [DDSP].			
Comments: This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).			
Category:	MDR	Priority:	Mandatory
Originator:	RAM	Origin:	
Target Release:	DIL_OBB_1.0.x	Scope:	System
Allocation:	DIL, IPS, ADM, DFE	Verification:	< will be added in a later document edition >
See also: ==> [EL4-DIL-SSS-REQ-00321] Reliability - Error Reporting - DDSP			
Issues/Decisions:			
LH Text:			
PH Text:			

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EL4-DIL-SSS-REQ-00323 Reliability - Failure of a server		(Accepted)
A [failure] of one server shall not affect the operational availability of other servers.		
<i>Comments:</i>		
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i>
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, IPS	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>		
<i>PH Text:</i>		

EL4-DIL-SSS-REQ-00324 Reliability - Failure of a PDU		(Accepted)
A [failure] of a [PDU] shall not have operational consequences.		
<i>Comments:</i>		
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i> ==> [EL4 PH] [02_46]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, IPS, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	3.6.0-1 Ein Tausch ausgefallener/defekter Komponenten muss rückwirkungsfrei auf den Betrieb und die Zuverlässigkeit gegenüber dem Gesamtsystem DSTW sein.	
<i>PH Text:</i>	Der Tausch einer ausgefallenen HW-Komponente (korrektive Wartungsmaßnahme) wird rückwirkungsfrei auf den Betrieb durchgeführt unter Einhaltung der Zuverlässigkeit gegenüber dem Gesamtsystem DSTW.	

EL4-DIL-SSS-REQ-00325 Reliability - Failure of a UPS		(Accepted)
A [failure] of a [UPS] shall not have operational consequences.		
<i>Comments:</i>		
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i> ==> [EL4 PH] [02_46]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, IPS, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	3.6.0-1 Ein Tausch ausgefallener/defekter Komponenten muss rückwirkungsfrei auf den Betrieb und die Zuverlässigkeit gegenüber dem Gesamtsystem DSTW sein.	
<i>PH Text:</i>	Der Tausch einer ausgefallenen HW-Komponente (korrektive Wartungsmaßnahme) wird rückwirkungsfrei auf den Betrieb durchgeführt unter Einhaltung der Zuverlässigkeit gegenüber dem Gesamtsystem DSTW.	

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EL4-DIL-SSS-REQ-00326 Reliability - Failure of a Connection		(In negotiation)
A [failure] of one connection shall not affect the operational availability of the other connections.		
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]). == OPEN POINT == Context of Requirement not defined. Type of connection and components to be defined.		
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, VIL, IPS, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>	
<i>Issues/Decisions:</i>		
<i>LH Text:</i>		
<i>PH Text:</i>		

EL4-DIL-SSS-REQ-00328 Reliability - Technical single Error		(Accepted)
The occurrence of a single technical [error] in the [DIL] system shall not lead to any operational restrictions.		
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, VIL, IPS, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>	
<i>Issues/Decisions:</i>		
<i>LH Text:</i>		
<i>PH Text:</i>		

EL4-DIL-SSS-REQ-00329 Reliability - Diagnosis of Failure of open Network		(Accepted)
The [DIL] shall be able to diagnose [failure]s of the open network (category 2 or 3 according to [EN50159]) and the data terminals with the respective location reference.		
<i>Comments:</i> Regardless of who is responsible for network operation, a disruption must be diagnosable. The network status must be recognizable in order to assess availability. Without this transparency, network difficulties cannot be identified and therefore cannot be taken into account in RAM analyses. This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
<i>Category:</i> MDR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, IPS	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i> ==> [EL4-DIL-SSS-REQ-00321] Reliability - Error Reporting - DDSP	
<i>Issues/Decisions:</i>		
<i>LH Text:</i>		
<i>PH Text:</i>		

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EL4-DIL-SSS-REQ-00330 Reliability - Diagnosis of failure of closed Network		(In negotiation)
The [DIL] shall be able to diagnose [failure]s of the closed network (category 1 according to [EN50159]) and the data terminals with the respective location reference.		
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]). == OPEN POINT == Clarification necessary to which extent a diagnosis is possible.		
<i>Category:</i> MDR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, IPS, ADM	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i> ==> [EL4-DIL-SSS-REQ-00321] Reliability - Error Reporting - DDSP	
<i>Issues/Decisions:</i>		
<i>LH Text:</i>		
<i>PH Text:</i>		

3.1.3 MAINTAINABILITY AND MAINTENANCE

3.1.3.1 DEFINITIONS

Maintainability:

The ability to remain in, or to be returned to, a functional state, under given application and maintenance conditions (definition according to [OEVE/OENORM EN 50126]).

Maintenance:

Generic term for inspection, servicing and repair.

3.1.3.2 GENERAL

EL4-DIL-SSS-REQ-00331 Maintainability and Maintenance - DDSP Compliance concerning Maintainability		(Accepted)
The DIL system shall comply with the [DDSP] concept to ensure maintainability.		
<i>Comments:</i>		
<i>Category:</i> DOR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_2.0.0 DCA_OBB_3.1.2 <i>Allocation:</i> DIL, IPS, ADM, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_53] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>	
<i>Issues/Decisions:</i>		
<i>LH Text:</i> 4.3.0-1 Für die Einhaltung der Instandhaltbarkeit muss das Konzept DDSP vom AN eingehalten werden.		
<i>PH Text:</i> Die konkreten Punkte sind bei Vorliegen einer detaillierten DDSP Spezifikation und Architektur gemeinsam festzulegen. Das derzeit vorliegende Konzept stellt keine ausreichende Spezifikation dar.		

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EL4-DIL-SSS-REQ-00332 Maintainability and Maintenance - IT Commissioning Compliance (Accepted)	
The DIL system shall comply with the [IT] commissioning concept to ensure maintainability.	
Comments: This refers to the [LH], part D9 "IT-Inbetriebnahme Konzept", and part D8 "Security Konzept".	
Category: DOR Originator: Customer Target Release: DIL_OBB_1.0.x DCA_OBB_3.1.0 Allocation: DIL, VIL, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_54] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text: 4.3.0-2 Für die Einhaltung der Instandhaltbarkeit muss das IT Inbetriebnahmekonzept vom AN eingehalten werden.	
PH Text: Die Einhaltung des IT Inbetriebnahmekonzepts wird vom AN im IT Security Konzept dargelegt.	

EL4-DIL-SSS-REQ-00333 Maintainability and Maintenance - EULYNX Compliance (Accepted)	
The DIL system shall comply with the [EULYNX specification] - (Eu.Doc.18) to ensure maintainability.	
Comments: SMI part is scope of DCA 3.1.2.	
Category: DOR Originator: Customer Target Release: DIL_OBB_1.0.x DCA_OBB_3.1.0 Allocation: DIL, VIL, ADM, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_55] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text: 4.3.0-3 Für die Einhaltung der Instandhaltbarkeit müssen die relevanten Dokumente der EULYNX-Spezifikation vom AN eingehalten werden.	
PH Text: Für die Einhaltung der Instandhaltbarkeit wird der AN die EULYNX-Spezifikation Eu.Doc.18 Version: 4.0 (2.A) einhalten.	

3.1.3.3 STORAGE OF SPARE PARTS

EL4-DIL-SSS-REQ-00334 Maintainability and Maintenance - Spare Parts Storage - Concept (Accepted)	
The supplier shall develop and provide a storage concept for the spare parts as described in the service description.	
Comments: == MISMATCH == This is a requirement, but in the LH it is only marked as "Info".	
Category: MDR Originator: Customer Target Release: DIL_OBB_1.0.x Allocation: DIL, VIL, IPS, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_57] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text: 4.4.0-1 Der AN entwickelt ein Konzept der Lagerung für die Ersatzteile, wie in der Leistungsbeschreibung beschrieben.	
PH Text:	

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EL4-DIL-SSS-REQ-00335		Maintainability and Maintenance - Spare Parts Storage - Information		(Accepted)			
The supplier shall provide any information and facilities for spare parts storage.							
Comments:							
Category:		MDR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_58]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL, IPS, DFE		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		4.4.0-2 Allfällige Angaben und Einrichtungen zur Ersatzteillagerung müssen vom AN bereit gestellt werden. Grundsätzlich sind Komponenten so zu entwickeln, dass eine Lagerhaltung mit minimalem Aufwand möglich ist.					
PH Text:		Angaben und Einrichtungen zur Ersatzteillagerung werden vom AN bereitgestellt. Die Komponenten werden so entwickelt, dass eine Lagerhaltung mit minimalem Aufwand möglich ist.					

EL4-DIL-SSS-REQ-00336		Maintainability and Maintenance - Spare Parts		(In negotiation)	
		Storage - Effort			
Components shall be developed in such a way that storage is possible with minimal effort.					
Comments:		== OPEN POINT ==			
		Requirement has to be refined according to EN50125-3 oder EN 60721-3.			
Category:	MDR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_58]		
Target Release:	DIL_OBB_1.0.x	Scope:	System		
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:		4.4.0-2 Allfällige Angaben und Einrichtungen zur Ersatzteillagerung müssen vom AN bereit gestellt werden. Grundsätzlich sind Komponenten so zu entwickeln, dass eine Lagerhaltung mit minimalem Aufwand möglich ist.			
PH Text:		Angaben und Einrichtungen zur Ersatzteillagerung werden vom AN bereitgestellt. Die Komponenten werden so entwickelt, dass eine Lagerhaltung mit minimalem Aufwand möglich ist.			

EL4-DIL-SSS-REQ-00337		Maintainability and Maintenance - Spare Parts Storage - Maintenance		(Accepted)	
All spare parts shall be storable without requiring regular maintenance and without being kept under voltage.					
Comments:		Exceptions for individual components require an alternative solution from the contractor and shall be approved by the customer.			
Category:	MDR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_59]		
Target Release:	DIL_OBB_1.0.x	Scope:	System		
Allocation:	DCA_OBB_3.1.0	Verification:	< will be added in a later document edition >		
	DIL, IPS, DFE	See also:			
Issues/Decisions:					
LH Text:	4.4.0-3 Alle Ersatzteile müssen lagerfähig sein, ohne dass sie einer regelmäßigen Instandhaltung unterliegen sowie unter Spannung gehalten werden. Ausnahmen für einzelne Komponenten erfordern eine alternative Lösung des AN und müssen vom AG genehmigt werden.				
PH Text:	Alle Ersatzteile sind lagerfähig, ohne dass sie einer regelmäßigen Instandhaltung unterliegen sowie unter Spannung gehalten werden müssen.				

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EL4-DIL-SSS-REQ-00338		Maintainability and Maintenance - Spare Parts Storage - Climate Class		(Accepted)
Spare parts shall be storable under the environmental conditions of Climate Class T2 according to [SSS_DIL_GENERAL], chapter "Environmental Conditions".				
Comments:				
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_60]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	4.4.0-4 Ersatzteile müssen unter den Umgebungsbedingungen gemäß den Klimaklassen in dem <03 Lastenheft Allgemeine Anforderungen> 1 lagerfähig sein.			
PH Text:	Die Umgebungsbedingungen der Ersatzteile können variieren, in Abhängigkeit zu dem geplanten Einsatzort- und Zweck (Feld, FES, Rechenzentrum, etc.). Sensible Ersatzteile sind ggf. nur unter den Umgebungsbedingungen gemäß der Klimaklasse Anstrich 6. "im Gebäude klimatisiert: +18 °C bis +27 °C" lagerfähig.			

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3.1.3.4 LABELLING

EL4-DIL-SSS-REQ-00339 Maintainability and Maintenance - Labelling - Data (In negotiation)	
All hardware components and modules of the indoor and outdoor systems, including switch cabinets and switch boxes, shall be clearly labelled in machine-readable form using a [QR] code and shall show at least the following data: → Manufacturer → Type → Item number → Device version (hardware and firmware version, etc.) → Serial number → [MAC] addresses	
Comments: Hitachi Rail applies the GS1 data matrix code (2D barcode). == OPEN POINT == The customer shall specify the concrete form with the supplier during the specification phase.	
Category: MDR Originator: Customer Target Release: DIL_OBB_1.0.x DCA_OBB_3.1.0 Allocation: DIL, IPS, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_62] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text:	4.5.0-1 Sämtliche Komponenten, Module der Innen- und Außenanlage inkl. Schaltschränke und Schaltkästen müssen in maschinenlesbarer Form mittels QR-Code eindeutig gekennzeichnet sein und mindestens folgende Daten aufweisen: • Hersteller • Type • Sachnummer • Geräteausgabestand (Hardware-, Firmwareversion, etc.) • Seriennummer • MAC-Adressen Die konkrete Ausformung ist im Rahmen des Pflichtenheftphase mit dem AG abzustimmen.
PH Text:	Hitachi Rail wendet den GS1-DataMatrixcode an. Hitachi Rail – Logo, Feld mit 2D Barcode (01) GS1 Nummer (21) Seriennummer (240) Sachnummer (7021) PCS (Ausgabestand) (zusätzlich Ausgabe der Schaltung) Bezeichnung

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EL4-DIL-SSS-REQ-00340		Maintainability and Maintenance - Labelling - Units		(In negotiation)
Labelling shall be applied to the smallest replaceable unit, for example:				
→ Object Controller				
→ Power supply units				
→ Light point				
→ Wheel sensor				
→ Lightning protection				
→ Signal box				
→ Separately mountable parts by the supplier				
→ [FES]				
→ Server				
Comments:				
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_63]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
	DCA_OBB_3.1.0			
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	4.5.0-2 Die Kennzeichnung erfolgt auf die kleinste Tauschbare Einheit, Liste beispielhaft:			
	• Object Controller			
	• StrV (Netzteile)			
	• Lichtpunkt			
	• Radsensor			
	• Blitzschutz			
	• Signalbeikasten			
	• separat montierbare Teile durch AN			
	• FES			
	• Server			
	Details müssen vom AN mit dem AG während der Pflichtenheftphase definiert werden.			
PH Text:	Die Kennzeichnung erfolgt auf die kleinste Tauschbare Einheit.			

EL4-DIL-SSS-REQ-00341		Maintainability and Maintenance - Labelling - QR-Code for Outdoor System		(Accepted)
The [QR] code for the outdoor system shall be weatherproof and installed in a weather-resistant way.				
Comments:				
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_64]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	VIL, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	4.5.0-3 Der QR-Code für Außenanlage muss wetterfest/witterungsfest und entsprechend montiert sein.			
PH Text:	Das Typenschild mit GS1-DataMatrixcode wird wetterfest/witterungsfest angebracht.			

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EL4-DIL-SSS-REQ-00342		Maintainability and Maintenance - Labelling - QR-Code - Format		(In negotiation)			
The supplier shall comply with the (QR) code standard GS1-DataMatrixcode and data format.							
Comments: == OPEN POINT == GS1-DataMatrixcode to be approved by customer.							
Category:		MDR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_65]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL, IPS, DFE		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		4.5.0-4 Der AN muss den QR-Code Standard bzw. Datenformat einhalten.					
PH Text:		Hitachi Rail wendet den GS1-DataMatrixcode an.					

EL4-DIL-SSS-REQ-00343		Maintainability and Maintenance - Labelling - Serial Number		(Accepted)	
The serial number of the components in the spare parts inventory shall enable a unique identification of the components throughout their entire operating life.					
Comments:		The serial numbers of the components allow for transparent tracking of components throughout their entire operating life (e.g. in [DDSP], [AVS], [SAP]).			
Category:	MDR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_66] ==> [EL4 PH] [02_67]		
Target Release:	DIL_OBB_1.0.x DCA_OBB_3.1.0	Scope:	System		
Allocation:	DIL, IPS, DFE	Verification:	< will be added in a later document edition >		
Issues/Decisions:		See also:			
LH Text:		4.5.0-5 Die Seriennummer der Komponenten der Ersatzteilhaltung muss eine eindeutige Identifikation der Komponenten über die gesamte Nutzungsdauer ermöglichen. 4.5.0-6 Die Seriennummer der Komponenten dient der transparenten Verfolgung der Komponenten über die gesamte Nutzungsdauer (z.B. im DDSP, AVS, SAP).			
PH Text:		Über die Seriennummer ist eine eindeutige Identifikation möglich.			

3.1.3.5 MAINTENANCE DOCUMENTATION

Maintenance is carried out in accordance with the service description and service level agreement.

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EL4-DIL-SSS-REQ-00345 Maintainability and Maintenance - Documentation - Usability (Accepted)	
The supplier shall deliver the maintenance documentation in a form that enables the customer to perform the inspection, maintenance and repair (at least component replacement, troubleshooting, resetting, restarting, etc.) of the supplied system.	
Comments:	
Category: DOR Originator: Customer Target Release: DIL_OBB_1.0.x Allocation: DIL, VIL, IPS, ADM, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_70] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text:	4.6.0-2 Die Instandhaltungsdokumentation muss die Inspektion, Wartung und Instandsetzung (mindestens Komponententausch, Fehlersuche, Reset, Neustart, usw.) der vom AN gelieferten Anlagen durch den AG ermöglichen.
PH Text:	Die Instandhaltungsdokumentation wird die Inspektion, Wartung und Instandsetzung der vom AN gelieferten Anlagen ermöglichen. Dies enthält: notwendigen Komponententausch, Fehlersuche, Reset und Neustart.

EL4-DIL-SSS-REQ-00346 Maintainability and Maintenance - Documentation - Generic Form (Accepted)	
The supplier shall prepare the maintenance documentation generically, so that it is applicable to all sites of the supplied system.	
Comments: <u>Note:</u> The preparation of a site specific maintenance documentation by the supplier is not planned.	
Category: DOR Originator: Customer Target Release: DIL_OBB_1.0.x Allocation: DIL, VIL, IPS, ADM, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_71] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text:	4.6.0-3 Die Instandhaltungsdokumentation muss generisch erstellt und für sämtliche vom AN gelieferten Anlagen anwendbar sein. 4.6.0-4 Es ist nicht vorgesehen, dass der AN eine anlagenspezifische Instandhaltungsdokumentation liefert.
PH Text:	Die Instandhaltungsdokumentation wird eine generische Struktur aufweisen. Der AN erstellt und übergibt für die von ihm gelieferten Produkte Instandhaltungsdokumente.

EL4-DIL-SSS-REQ-00347 Maintainability and Maintenance - Documentation - Electronic Form (Accepted)	
The supplier shall provide the maintenance documentation in printed and electronic form and make it available for the [DDSP].	
Comments: Note: The documentation is transferred to the customer.	
Category: DOR Originator: Customer Target Release: DIL_OBB_1.0.x Allocation: DIL, VIL, IPS, ADM, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_73] Scope: System Verification: < will be added in a later document edition > See also:
Issues/Decisions:	
LH Text:	4.6.0-5 Die Instandhaltungsdokumentation muss gedruckt, elektronisch und für das DDSP zur Verfügung gestellt werden.
PH Text:	Die Instandhaltungsdokumentation wird in gedruckter Form und elektronisch an den AG übergeben.

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EL4-DIL-SSS-REQ-00348 Maintainability and Maintenance - Documentation - File Format		(Accepted)
The file format for the maintenance documentation in [DDSP] shall be agreed between the customer and the supplier during the specification phase.		
<i>Comments:</i> Proposal for standardized file formats: .docx and .pdf.		
<i>Category:</i>	DOR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_74]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, VIL, IPS, ADM, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	4.6.0-6 Das Dateiformat für die Instandhaltungsdokumentation im DDSP ist während der Pflichtenheftphase zwischen AG und AN abzustimmen.	
<i>PH Text:</i>	Der AN schlägt die standardisierten Dateiformate .docx und .pdf vor.	

3.1.3.6 MAINTENANCE PROCESS

EL4-DIL-SSS-REQ-00349 Maintainability and Maintenance - Process - Implementation and Practice		(Accepted)
The supplier shall implement a maintenance process together with the customer and support the the customer during its execution.		
<i>Comments:</i>		
<i>Category:</i>	GCR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_76]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	4.7.0-1 Es muss ein Instandhaltungsprozess zwischen AN und AG implementiert und gelebt werden. Systeme, Teilsysteme und Komponenten, die vom AN geliefert werden, müssen unter Berücksichtigung des Konzept DDSP dem Instandhaltungsprozess entsprechen.	
<i>PH Text:</i>	Betriebsführung und Instandhaltung wird vom AG durchgeführt. Prozesse zu incident, change und operations management werden somit vom AG umgesetzt und durchgeführt. Der AN unterstützt dabei den AG als 2. und 3. Level Support, der 1. Level liegt allein in der Verantwortung des AG. Der AN ist vom Auftraggeber über die Prozesse in Kenntnis zu setzen die den 2. und 3. Level Support betreffen, damit die Servicecenter entsprechend ihre Prozesse darauf ausrichten können. Der AN wird entsprechende Wartungs- und Instandhaltungsdokumente zur Verfügung stellen.	

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EL4-DIL-SSS-REQ-00817 Maintainability and Maintenance - Process - DDSP Compliance concerning Maintenance		(Accepted)
The supplier shall ensure that the supplied systems, subsystems and components comply with the maintenance process, taking into account the DDSP concept.		
<i>Comments:</i>		
<i>Category:</i>	GCR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_76]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	4.7.0-1 Es muss ein Instandhaltungsprozess zwischen AN und AG implementiert und gelebt werden. Systeme, Teilsysteme und Komponenten, die vom AN geliefert werden, müssen unter Berücksichtigung des Konzept DDSP dem Instandhaltungsprozess entsprechen.	
<i>PH Text:</i>	Betriebsführung und Instandhaltung wird vom AG durchgeführt. Prozesse zu incident, change und operations management werden somit vom AG umgesetzt und durchgeführt. Der AN unterstützt dabei den AG als 2. und 3. Level Support, der 1. Level liegt allein in der Verantwortung des AG. Der AN ist vom Auftraggeber über die Prozesse in Kenntnis zu setzen die den 2. und 3. Level Support betreffen, damit die Servicecenter entsprechend ihre Prozesse darauf ausrichten können. Der AN wird entsprechende Wartungs- und Instandhaltungsdokumente zur Verfügung stellen.	

EL4-DIL-SSS-REQ-00350 Maintainability and Maintenance - Process - Access		(In negotiation)
Maintenance staff shall be able to access the relevant service documentation (locally or from the DDSP) by using the (QR code on the individual elements (e.g. OC).		
<i>Comments:</i>		
== MISMATCH ==		
Das ist eine Anforderung, aber im LH nur als "Info" markiert.		
== OPEN POINT ==		
GS1 Matrix Code only contains Info about HW (e.g. part number) no hyperlink.		
Maintaining a direct hyperlink over lifetime is impossible.		
<i>Category:</i>	MDR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_77]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	4.7.0-2 Mittels QR-Code an den einzelnen Elementen (z.B. OC) soll der Instandhalter die dazugehörige Service-Dokumentation aufrufen können (lokal oder vom DDSP).	
<i>PH Text:</i>		

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EL4-DIL-SSS-REQ-00351 Maintainability and Maintenance - Process - Automatic Earth Short Circuit Measurements		(Accepted)
During operation only automatic earth short circuit measurements shall be permitted.		
<i>Comments:</i>		
<i>Category:</i>	MDR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_90]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	4.9.0-4 Im Betrieb sind keine manuellen Erdschlussmessungen zugelassen.	
<i>PH Text:</i>	In der FES ist eine automatische Erdschlussüberwachung vorgesehen. Manuelle Erdschlussmessungen sind somit im Betrieb nicht erforderlich.	

3.1.3.7 COMPONENT REPLACEMENT

EL4-DIL-SSS-REQ-00352 Maintainability and Maintenance - Component Replacement - Restrictions		(Accepted)
It shall not be possible for the maintenance stuff to locally reconfigure modules or change the data preparation of components that have been replaced due to [fault]s or [error]s.		
<i>Comments:</i>		
<i>Category:</i>	MDR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_79]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, VIL, IPS, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	4.8.0-1 Eine Neukonfiguration oder Umprojektierung von Modulen darf bei fehler- und störungsbedingtem Austausch durch das Instandhaltungspersonal vor Ort nicht möglich sein.	
<i>PH Text:</i>	Eine Neukonfiguration oder Umprojektierung ist nur mit entsprechender Projektierungssoftware und unter Einhaltung der Instandhaltungsdokumentation möglich.	

EL4-DIL-SSS-REQ-00353 Maintainability and Maintenance - Component Replacement - OC Replacement - Restrictions		(Accepted)
The [OC]s shall be designed in such a way that replacement with an incorrect spare part is not possible.		
<i>Comments:</i>	If it is not possible to distinguish between different [OC]s using hardware, the wrong [OC] shall be detected by software. A status error with [LED] must be displayed on the device.	
<i>Category:</i>	MDR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_80]
<i>Target Release:</i>	DIL_OBB_1.0.x DCA_OBB_3.1.0	<i>Scope:</i> System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	4.8.0-2 Object Controller müssen so konstruiert sein, dass der Tausch mit einem falschen Ersatzteil nicht möglich ist.	
<i>PH Text:</i>	Die von Hitachi Rail gelieferten Object Controller sind nur mit dem passenden Backpanel kompatibel. Ein Tausch mit einem falschen Ersatzteil eines anderen Lieferanten wird daher ausgeschlossen. Die Object Controller Typen LS und IO haben keinen konstruktiven Schutz gegen Austauschbarkeit untereinander. Allerdings wird auf Protokoll Ebene ein falscher Object Controller Typ erkannt.	

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EL4-DIL-SSS-REQ-00354 Maintainability and Maintenance - Component Replacement - OC Replacement - Impact (Accepted)	
The replacement of an [OC] shall not result in any operational effects or [fault]s other than the replacement of this [OC].	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DCA_OBB_3.1.0 <i>Allocation:</i> DIL, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_81] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	4.8.0-3 Der Tausch eines Object Controllers darf über den Tausch dieses Object Controllers hinaus zu keiner weiteren Beeinflussung des Betriebs oder Störung führen.
<i>PH Text:</i>	Der Tausch eines Object Controllers oder eines seiner Teilkomponenten ist in seinen Auswirkungen lokal begrenzt auf die betroffenen Elemente des Object Controllers.

EL4-DIL-SSS-REQ-00355 Maintainability and Maintenance - Component Replacement - Approved Tools (In negotiation)	
The assembly, disassembly and repair of components shall be carried out with tools that have been approved by the customer.	
<i>Comments:</i> == OPEN POINT == Tools to be agreed with customer during design phase.	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, IPS, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_82] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	4.8.0-4 Die Montage, Demontage und Instandsetzung von Komponenten muss mit Werkzeugen ausgeführt werden, die vom AG zu genehmigen sind.
<i>PH Text:</i>	Der AN wird die notwendigen Werkzeuge vorab mit dem AG abstimmen und anschließend in den Instandhaltungsdokumenten anführen.

EL4-DIL-SSS-REQ-00356 Maintainability and Maintenance - Component Replacement - Standard Tools (Accepted)	
No specialized tool (designed to work only with the component being replaced) shall be required to replace any component.	
<i>Comments:</i>	
<i>Category:</i> MDR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DCA_OBB_3.1.0 <i>Allocation:</i> DIL, IPS, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_83] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	4.8.0-5 Für den Tausch von Komponenten darf kein Spezialwerkzeug (welches nur für den Einsatzbereich dieser Komponente entwickelt wurde) notwendig sein.
<i>PH Text:</i>	Der AN plant keinen Einsatz von Spezialwerkzeugen für die Instandhaltung seiner Komponenten und Systeme.

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EL4-DIL-SSS-REQ-00357 Maintainability and Maintenance - Component Replacement - Manual corrective Maintenance Actions		(Accepted)
When replacing components during maintenance, the required (MRT) value according to chapter "[D2.2 RAMS/General/Assumptions]" shall be maintained.		
Comments: Assumption 4: Concerning the availability calculation for the manual repair of a [fault] an [MRT] of 0,5h is assumed. Assumption 5: Concerning the availability calculation for the manual repair of an [error] an [MRT] of 1h is assumed. The times for replacing components shall be checked and documented as examples during project implementation.		
Category:	MDR	Priority: Mandatory
Originator:	Customer Technics (R&D)	Origin: ==> [EL4 PH] [02_84]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, IPS, DFE	Verification: < will be added in a later document edition > See also:
Issues/Decisions:		
LH Text:	4.8.0-6 Für den Tausch von Komponenten muss im Zuge der Instandhaltung nachweislich der geforderte MRT-Wert eingehalten werden.	
PH Text:	Die Zeiten für den Tausch von Komponenten wird im Zuge der Projektumsetzung exemplarisch geprüft und dokumentiert.	

EL4-DIL-SSS-REQ-00358 Maintainability and Maintenance - Component Replacement - Sufficent Data to DDSP		(In negotiation)
The [DIL-CU] shall deliver all the necessary data to the DDSP so that all required information for maintenance activities concerning component exchange can be derived from the DDSP.		
Comments: • The customer provides as software tool a data diagnosis service platfrom [DDSP]. • Avoiding additional apps/applications/software requires the integration of third components into the [DDSP]. == OPEN POINT == Specification of the interface between [DIL] and [DDSP]		
Category:	MDR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_85] ==> [EL4 PH] [02_86]
Target Release:	DIL_OBB_2.0.0	Scope: System
Allocation:	DIL, IPS	Verification: < will be added in a later document edition > See also:
Issues/Decisions:		
LH Text:	4.8.0-7 Als Softwarewerkzeug für die Instandhaltung stellt der AG eine Daten-Diagnose-Service-Plattform (DDSP) bei. 4.8.0-8 Für die Instandhaltung dürfen zur DDSP keine zusätzlichen Apps/Applikationen/Software benötigt werden.	
PH Text:	Instandhaltung kann über das DDSP ohne zusätzliche Apps/Applikationen/Software erfolgen. Dazu müssen Drittkomponenten im DDSP integriert werden.	

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3.1.3.8 MISCELLANEOUS

EL4-DIL-SSS-REQ-00359 Maintainability and Maintenance - No Inspection (Accepted) and Maintenance			
The following supplier-provided components shall be inspection- and maintenance-free: → (DC) components <u>Exceptions:</u> → Regular cleaning or replacement of filter mats and fans of network components If this cannot be achieved, a minimum maintenance interval of 6 months is allowed.			
<i>Comments:</i>			
<i>Category:</i>	MDR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_88]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i>	System
<i>Allocation:</i>	DIL, IPS	<i>Verification:</i>	< will be added in a later document edition >
<i>See also:</i>			
<i>Issues/Decisions:</i>			
<i>LH Text:</i>	4.9.0-1 Alle Komponenten des Rechenzentrums, die vom AN geliefert werden, müssen inspektions- und wartungsfrei sein.		
<i>PH Text:</i>	Die vom AN gelieferten Server sind Wartungs- und Inspektionsfrei.		

EL4-DIL-SSS-REQ-01479 Maintainability and Maintenance - No Inspection (Accepted) and Maintenance			
The following supplier-provided components shall be inspection- and maintenance-free: → Field elements (including their stations) and OCs <u>Exceptions:</u> → Cleaning of the signal optics of the light signal subsystem → Adjusting of the axle counter of the subsystem (TDS). → Regular cleaning or replacement of filter mats and fans of network components If this cannot be achieved, a minimum maintenance interval of 6 months is allowed.			
<i>Comments:</i>			
<i>Category:</i>	MDR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_89] ==> [EL4 PH] [02_258]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i>	System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i>	< will be added in a later document edition >
<i>See also:</i>			
<i>Issues/Decisions:</i>			
<i>LH Text:</i>	4.9.0-2 Alle Feldelemente müssen inspektionsfrei sein und dürfen keine regelmäßige Wartungsmaßnahmen erfordern. Ausgenommen ist die Reinigung der Signaloptik des Subsystems Lichtsignal und die Justierung des Achszählers des Subsystems Gleisfreimeldung. 4.9.0-3 Feldelementstationen und Object Controller müssen wartungsfrei sein.		
<i>PH Text:</i>	Die vom AN gelieferten Feldelemente sind Wartungs- und Inspektionsfrei. Die vom AN gelieferten Teile der Feldelementstationen sowie die Object Controller sind Wartungs- und Inspektionsfrei.		

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EL4-DIL-SSS-REQ-00360 Maintainability and Maintenance - No preventive Maintenance (Accepted)	
The [DIL] shall be free of preventive maintenance activities. If this is not technologically feasible for components or if the performance or service life is increased as a result, a maximum of a half-year period maintenance interval is granted.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> MDR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, IPS, ADM, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

EL4-DIL-SSS-REQ-00361 Maintainability and Maintenance - Easy Accessibility (Accepted)	
Servers and cables shall be easily accessible for maintenance. Therefore, the servers shall be able to be pulled out fully during operation and cable management shall be done with cable management arms.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> MDR <i>Originator:</i> Technics (R&D) <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, IPS	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

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3.1.3.9 EMPLOYEE PROTECTION

EL4-DIL-SSS-REQ-00566 Maintainability and Maintenance - Employee Protection - Product Safety (Accepted)	
The supplier shall comply with the requirements concerning product safety according to [EN 50122], [EN 61010-1] and [EN 62368-1] to ensure employee protection.	
Components, subsystems and systems supplied by the supplier shall be designed in such a way that the employee protection is ensured when the customer carries out maintenance works.	
Comments: The employee protection for the customer's employees can only be ensured by the customer himself.	
Category:	GCR
Originator:	Customer
Target Release:	DIL_OBB_1.0.x
Allocation:	DCA_OBB_3.1.0 DIL, IPS, DFE
Priority:	Mandatory
Origin:	==> [EL4 PH] [02_92]
Scope:	System
Verification:	< will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	4.10.0-1 Komponenten, Teilsysteme und Systeme, welche durch den AN geliefert werden, müssen so konzipiert sein, damit bei der Durchführung der Instandhaltungstätigkeiten durch den AG der Arbeitnehmerschutz sichergestellt wird.
PH Text:	Der AN wird die folgenden Anforderungen bezüglich Produktsicherheit berücksichtigen: EN 50122 EN 61010-1 EN 62368-1 Bezüglich Arbeitnehmerschutz wird die R10 (Eisenbahnanlagen Schwerpunkt-konzept aus Sicht des Arbeitnehmer/innenschutzes) herangezogen werden.

EL4-DIL-SSS-REQ-01476 Maintainability and Maintenance - Employee Protection - R10 (Accepted)	
The supplier shall comply with the regulation [R10] to ensure employee protection.	
Comments: The employee protection for the customer's employees can only be ensured by the customer himself.	
Category:	GCR
Originator:	Customer
Target Release:	DIL_OBB_1.0.x
Allocation:	DIL, IPS, DFE
Priority:	Mandatory
Origin:	==> [EL4 PH] [02_92]
Scope:	System
Verification:	< will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	4.10.0-1 Komponenten, Teilsysteme und Systeme, welche durch den AN geliefert werden, müssen so konzipiert sein, damit bei der Durchführung der Instandhaltungstätigkeiten durch den AG der Arbeitnehmerschutz sichergestellt wird.
PH Text:	Der AN wird die folgenden Anforderungen bezüglich Produktsicherheit berücksichtigen: EN 50122 EN 61010-1 EN 62368-1 Bezüglich Arbeitnehmerschutz wird die R10 (Eisenbahnanlagen Schwerpunkt-konzept aus Sicht des Arbeitnehmer/innenschutzes) herangezogen werden.

3.1.4 AVAILABILITY

3.1.4.1 DEFINITIONS

Availability:

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The ability to perform a required function at a given time or during a given time interval under given conditions, provided that the necessary external aids are provided (definition according to [OEVE/OENORM EN 50126]).

Automatic correction:

In the context of a "safe computing unit" (e.g. a [VIL Node]) means the following:

After a [fault] has been detected, a safe computing unit enters a safe state and performs a self-test. After a successful self-test, the disturbed module is automatically recommissioned.

This also applies to the other availability requirements in this chapter.

3.1.4.2 AVAILABILITY CLASSES

Depending on the required availability values, the following availability classes are defined:

- Class 1: Availability is 0.99999
- Class 2: Availability is 0.99995
- Class 3: Availability is 0.9999
- [Class 4: not available]
- Class 5: 0,9998

The availability values were determined in accordance with [OEVE/OENORM EN 50126] in a top-down approach as follows:

$$A = [MTTF]/([MTTF]+[MTTR])$$

3.1.4.3 DIL-CU AVAILABILITY

According to [Figure 1: System Overview] the [DIL-CU] comprises the subsystems [VIL], [ADM] and [IPS]. [IPS] also provides the necessary server hardware and network components to run them. The availability is defined from an architecture-independent perspective.

3.1.4.3.1 GENERAL

EL4-DIL-SSS-REQ-00406 Availability - [DIL-CU] - Overall		(Accepted)
The [DIL-CU] shall have a class 1 overall availability of at least 0,99999.		
Comments:	System, Subsystem outages with large-scale operational impact (Availability class 1 corresponds defect class FK1). Applicable and valid for basic virtualisation and georedundancy.	
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_104] ==> [EL4 PH] [02_106]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, VIL, IPS	Verification: < will be added in a later document edition > See also:
Issues/Decisions:		
LH Text:	5.3.0-5 Die Verfügbarkeit der DSTW-ZE muss mindestens 0,99999 betragen. 5.3.0-7 Jede DSTW-ZE wird der Verfügbarkeitsklasse 1 zugeordnet.	
PH Text:	Für die DSTW-ZE kommt eine hochverfügbare Server- und Netzwerkarchitektur in einer hot-hot Redundanz zum Einsatz. Diese Charakteristik ist unabhängig von einer Georedundanz gegeben. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden. Die Hochverfügbarkeit der virtualisierten Hardwarekomponenten wird über so genannte Verfügbarkeitszonen (engl. „availability zones“) gewährleistet.	

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EL4-DIL-SSS-REQ-00407		Availability distribution of [DIL-CU]		(Accepted)	
The distribution of the availability of the entire [DIL-CU] must correspond to class 1. This means in particular an availability of:					
0,999998 % (MTTR = 6 h)					
0,999996 % (MTTR = 10 min)					
0,999996 % (MTTR = 2 min)					
Comments:		Equal availabilities with different MTTR values are derived from customer requirements stating that outages with MTTR = 2 min may occur more often per N years than outages with MTTR = 10 min.			
Category:	DPR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_107]		
			==> [EL4 PH] [02_109]		
			==> [EL4 PH] [02_110]		
Target Release:	DIL_OBB_1.0.x	Scope:	System		
Allocation:	DIL, VIL, IPS	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:	5.3.0-8 Die Verfügbarkeit der DSTW-ZE in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 1) muss mindestens 0,999998 betragen. 5.3.0-10 Die Verfügbarkeit der DSTW-ZE in Bezug auf Störungen, welche durch die DSTW-ZE innerhalb von 10 Minuten automatisch selbst behoben werden (siehe Tabelle 1), muss mindestens 0,999996 betragen. 5.3.0-11 Die Verfügbarkeit der DSTW-ZE in Bezug auf Störungen, welche durch die DSTW-ZE innerhalb von 2 Minuten automatisch selbst behoben werden (siehe Tabelle 1), muss mindestens 0,999996 betragen.				
PH Text:	Für die DSTW-ZE kommt eine hochverfügbare Server- und Netzwerkarchitektur in einer hot-hot Redundanz zum Einsatz. Diese Charakteristik ist unabhängig von einer Georedundanz gegeben. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden. Die Hochverfügbarkeit der virtualisierten Hardwarekomponenten wird über so genannte Verfügbarkeitszonen (engl. „availability zones“) gewährleistet. Für die DSTW-ZE kommt eine hochverfügbare Server- und Netzwerkarchitektur in einer hot-hot Redundanz zum Einsatz. Diese Charakteristik ist unabhängig von einer Georedundanz gegeben. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden. Die Hochverfügbarkeit der virtualisierten Hardwarekomponenten wird über so genannte Verfügbarkeitszonen (engl. „availability zones“) gewährleistet. Für die DSTW-ZE kommt eine hochverfügbare Server- und Netzwerkarchitektur in einer hot-hot Redundanz zum Einsatz. Diese Charakteristik ist unabhängig von einer Georedundanz gegeben. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden. Die Hochverfügbarkeit der virtualisierten Hardwarekomponenten wird über so genannte Verfügbarkeitszonen (engl. „availability zones“) gewährleistet.				

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EL4-DIL-SSS-REQ-00408 Availability - [DIL-CU] - Outages with no operational Impact		(Accepted)
Outages of a [DIL-CU] with no operational impact shall have an availability of at least 0,997154.		
Comments: Outages of a [DIL-CU] with no operational impact correspond to defect class FK4.		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_108]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, VIL, IPS	Verification: < will be added in a later document edition >
See also:		
Issues/Decisions:		
LH Text:	5.3.0-9 Die Verfügbarkeit der DSTW-ZE in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben (siehe Tabelle 1), muss mindestens 0,997154 betragen.	
PH Text:	Für die DSTW-ZE kommt eine hochverfügbare Server- und Netzwerkarchitektur in einer hot-hot Redundanz zum Einsatz. Diese Charakteristik ist unabhängig von einer Georedundanz gegeben. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden. Die Hochverfügbarkeit der virtualisierten Hardwarekomponenten wird über so genannte Verfügbarkeitszonen (engl. „availability zones“) gewährleistet.	

EL4-DIL-SSS-REQ-00567 Availability - DIL-CU - Combined SW Package		(Accepted)
In order to be able to comply with the availability classes of the [DIL-CU] (see chapter "[Availability Classes]"), the supplier shall provide the operating system and the operational software.		
Comments:		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_101]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, VIL, IPS, ADM	Verification: < will be added in a later document edition >
See also:		
Issues/Decisions:		
LH Text:	5.3.0-2 Um die Verfügbarkeiten der DSTW-ZE einhalten zu können, muss der AN das Betriebssystem und die funktionsbezogene Software zur Verfügung stellen.	
PH Text:	siehe Kommentar.	

3.1.4.3.2 VIL AVAILABILITY

3.1.4.3.2.1 GENERAL

EL4-DIL-SSS-REQ-00587 Availability - [VIL-Instance] - Overall		(Accepted)
The [VIL-Instance] shall have a class 2 overall availability of at least 0,99995.		
Comments:		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_113]
Target Release:	DIL_OBB_1.0.x	Origin: ==> [EL4 PH] [02_115]
Allocation:	DIL, VIL	Scope: System
		Verification: < will be added in a later document edition >
See also:		
Issues/Decisions:		
LH Text:	5.4.0-1 Die Verfügbarkeit des VSTW muss mindestens 0,99995 betragen. 5.4.0-3 Jedes VSTW wird der Verfügbarkeitsklasse 2 zugeordnet.	
PH Text:	Die virtuellen Stellwerke (VSTWs) werden in einer hot-hot Redundanz und gleichmäßig über die Availability Zones verteilt. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden.	

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EL4-DIL-SSS-REQ-00569 Availability distribution of [VIL-Instance]		(Accepted)
The distribution of the availability of a [VIL-Instance] must correspond to class 2. This means in particular an availability of		
0,999979 % (MTTR = 6 h)		
0,99999 % (MTTR = 10 min)		
0,999996 % (MTTR = 2 min)		
Comments: Equal availabilities with different MTTR values are derived from customer requirements stating that outages with MTTR = 2 min may occur more often per N years than outages with MTTR = 10 min.		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_116]
		==> [EL4 PH] [02_118]
		==> [EL4 PH] [02_119]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, VIL	Verification: < will be added in a later document edition >
		See also:
Issues/Decisions:		
LH Text:	5.4.0-4 Die Verfügbarkeit des VSTW in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 2) muss mindestens 0,999979 betragen. 5.4.0-6 Die Verfügbarkeit des VSTW in Bezug auf Störungen, welche durch das VSTW innerhalb von 10 Minuten automatisch selbst behoben werden (siehe Tabelle 2), muss mindestens 0,99999 betragen. 5.4.0-7 Die Verfügbarkeit des VSTW in Bezug auf Störungen, welche durch das VSTW innerhalb von 2 Minuten automatisch selbst behoben werden (siehe Tabelle 2), muss mindestens 0,999996 betragen.	
PH Text:	Die virtuellen Stellwerke (VSTWs) werden in einer hot-hot Redundanz und gleichmäßig über die Availability Zones verteilt. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden. Die virtuellen Stellwerke (VSTWs) werden in einer hot-hot Redundanz und gleichmäßig über die Availability Zones verteilt. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden. Die virtuellen Stellwerke (VSTWs) werden in einer hot-hot Redundanz und gleichmäßig über die Availability Zones verteilt. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden.	

EL4-DIL-SSS-REQ-00570 Availability - [VIL-Instance] - Outages with no operational Impact		(Accepted)
Outages of a [VIL-Instance] with no operational impact shall have an availability of at least 0,997154.		
Comments: Outages of a [VIL-Instance] with no operational impact correspond to defect class FK4.		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_117]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, VIL	Verification: < will be added in a later document edition >
		See also:
Issues/Decisions:		
LH Text:	5.4.0-5 Die Verfügbarkeit des VSTW in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben (siehe Tabelle 2), muss mindestens 0,997154 betragen.	
PH Text:	Die virtuellen Stellwerke (VSTWs) werden in einer hot-hot Redundanz und gleichmäßig über die Availability Zones verteilt. So wird sichergestellt, dass die geforderten Verfügbarkeitsanforderungen auch bei einem Serverausfall erfüllt werden.	

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EL4-DIL-SSS-REQ-00573		Availability - VIL - Minor SW/HW Change		(Accepted)
Minor changes of the [VIL] [SW] (= Patches) or the [HW] shall be carried out without operational restrictions.				
Comments:		• Example for [SW] Patch: Scurity update • See [SSS_DIL_GENERAL], chapter "Software Actualization".		
Category:	MCR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_120] ==> [EL4 PH] [02_152] ==> [EL4 PH] [02_260]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, VIL, IPS	Verification:	< will be added in a later document edition >	
See also:				
Issues/Decisions:				
LH Text:	5.4.0-8 Erweiterungen und/oder Änderungen der VSTW-Software (Software Updates/Upgrades, Projektierungs-, oder Konfigurationsänderungen), ausgenommen Erweiterungen und/oder Änderungen mit Safety-Relevanz, dürfen nicht zur Einschränkung des Betriebs unter die erlaubte Verfügbarkeit führen. 5.9.0-1 Ein planmäßiges Servicesoftwareupgrade bzw. ein planmäßiger Hardwaretausch muss rückwirkungsfrei auf den Betrieb und die Verfügbarkeit gegenüber dem Gesamtsystem DSTW sein. Der Bahnbetrieb muss weiterhin unterbrechungsfrei gewährleistet sein und es dürfen keine betrieblichen Ausfallzeiten entstehen. 5.4.0-10 Während der Durchführung eines Tauschs einer Software ohne Safety-Relevanz muss der Betrieb durch eine Redundanz übernommen werden.			
PH Text:	Softwareupdates werden in Betriebspausen durch die Aktivierung der Systemdaten durchführt. Siehe auch DDSP Use Case SysF4 „Systemdaten aktivieren“. Sperren und Hinweisschilder bleiben dabei wenn möglich erhalten. Die Betriebsunterbrechung ist im Normalfall die Bootzeit der betroffenen Komponente, sowie die Zeit für die Wiedereinbindung in das Gesamtsystem. Ausgenommen sind Erweiterungen und/oder Änderungen mit Safety-Relevanz. Sämtliche Änderungen müssen eine "Change Impact Analyse" durchlaufen und die Auswirkung auf das Gesamtsystem muss individuell unter Safety und §40 Begleitung betrachtet werden. Änderungen in der generischen Baseline, sowie Stationsdaten sind ohne Betriebsbeeinflussen jedoch zur Wahrung des Replika-Determinismus ausgeschlossen. Im technischen Konzept 9 hat der AN die Unterschiede zwischen Upgrades und Patches dargelegt, sowie die mögliche Auswirkung auf den Bahnbetrieb erörtert. Patches, die keine Änderungen in den sicherheitsrelevanten Teilen des VSTWs (Konfigurationsdaten) bedürfen, wie z.B. Security Patches, können in das VSTW eingebracht werden, ohne den Betrieb zu beeinflussen. Dazu wird ein Redundanzpartner des VSTWs außer Betrieb genommen, die neue SW aufgespielt und die Redundanz wieder in das System eingebunden. Mit den verbleibenden Redundanzen, die noch keinen Patch erhalten haben, wird anschließend analog verfahren.			

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EL4-DIL-SSS-REQ-00818		Availability - VIL - Major SW/HW Change		(Accepted)
Major changes of the [VIL] [SW], especially with safety impact, (= Upgrades) or the [HW] shall be carried out during operational breaks.				
Comments: • Examples for [SW] Upgrades: • Enhancement or correction of the application [SW] • Update of the data preparation • See [SSS_DIL_GENERAL], chapter "Software Actualization".				
Category: MCR Originator: Customer		Priority: Mandatory Origin: ==> [EL4 PH] [02_120] ==> [EL4 PH] [02_152] ==> [EL4 PH] [02_259]		
Target Release: DIL_OBB_1.0.x Allocation: DIL, VIL, IPS		Scope: System Verification: < will be added in a later document edition > See also:		
Issues/Decisions:				
LH Text: 5.4.0-8 Erweiterungen und/oder Änderungen der VSTW-Software (Software Updates/Upgrades, Projektierungs-, oder Konfigurationsänderungen), ausgenommen Erweiterungen und/oder Änderungen mit Safety-Relevanz, dürfen nicht zur Einschränkung des Betriebs unter die erlaubte Verfügbarkeit führen. 5.9.0-1 Ein planmäßiges Servicesoftwareupgrade bzw. ein planmäßiger Hardwaretausch muss rückwirkungsfrei auf den Betrieb und die Verfügbarkeit gegenüber dem Gesamtsystem DSTW sein. Der Bahnbetrieb muss weiterhin unterbrechungsfrei gewährleistet sein und es dürfen keine betrieblichen Ausfallzeiten entstehen. 5.4.0-9 Während der Durchführung eines Tauschs einer Software mit Safety-Relevanz soll der Betrieb durch eine Redundanz übernommen werden.				
PH Text: Softwareupdates werden in Betriebspausen durch die Aktivierung der Systemdaten durchführt. Siehe auch DDSP Use Case SysF4 „Systemdaten aktivieren“. Sperren und Hinweisschilder bleiben dabei wenn möglich erhalten. Die Betriebsunterbrechung ist im Normalfall die Bootzeit der betroffenen Komponente, sowie die Zeit für die Wiedereinbindung in das Gesamtsystem. Ausgenommen sind Erweiterungen und/oder Änderungen mit Safety-Relevanz. Sämtliche Änderungen müssen eine "Change Impact Analyse" durchlaufen und die Auswirkung auf das Gesamtsystem muss individuell unter Safety und §40 Begleitung betrachtet werden. Änderungen in der generischen Baseline, sowie Stationsdaten sind ohne Betriebsbeeinflussen jedoch zur Wahrung des Replika-Determinismus ausgeschlossen. Im technischen Konzept 9 hat der AN die Unterschiede zwischen Upgrades und Patches dargelegt, sowie die mögliche Auswirkung auf den Bahnbetrieb erörtert. Da es während der SW-Aktualisierung eine kurze Zeitspanne gibt, in der weder ein VSTW noch dessen Redundanzen operativ sind und daher keine Fahrwegsicherung mit freigestellten Signalen existiert, ist eine SW-Aktualisierung in einem geplanten Zeitintervall durchzuführen, in dem kein regulärer Zugverkehr stattfindet. Diese Zeitspanne ist mit 3-4 min begrenzt. Das ist jene Phase, ab der der FDL die "Zustimmung zur Aktivierung" gibt bis zum Zeitpunkt an dem das "VSTW-neu" die Generalabfrage zu seinen Nachbarsystemen und Außenanlagenelementen abgeschlossen hat, die betriebsrelevanten Daten wieder gesetzt sind und das VSTW wieder bedienbar ist.				

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EL4-DIL-SSS-REQ-00574		Availability - VIL - Operational Downtime		(Accepted)			
The operational downtime shall not be longer than the restart time of the affected component, as well as the time for the re-integration into the overall system.							
Comments:							
Category:		MCR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_120]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL, VIL, IPS		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		5.4.0-8 Erweiterungen und/oder Änderungen der VSTW-Software (Software Updates/Upgrades, Projektierungs-, oder Konfigurationsänderungen), ausgenommen Erweiterungen und/oder Änderungen mit Safety-Relevanz, dürfen nicht zur Einschränkung des Betriebs unter die erlaubte Verfügbarkeit führen.					
PH Text:		Softwareupdates werden in Betriebspausen durch die Aktivierung der Systemdaten durchführt. Siehe auch DDSP Use Case SysF4 „Systemdaten aktivieren“. Sperren und Hinweisschilder bleiben dabei wenn möglich erhalten. Die Betriebsunterbrechung ist im Normalfall die Bootzeit der betroffenen Komponente, sowie die Zeit für die Wiedereinbindung in das Gesamtsystem. Ausgenommen sind Erweiterungen und/oder Änderungen mit Safety-Relevanz.					

3.1.4.3.3 IPS AVAILABILITY

EL4-DIL-SSS-REQ-01480		Availability - IPS - Overall		(Accepted)	
The [IPS] shall have a class 1 overall availability of at least 0,99999.					
Comments:					
Category:		DPR		Priority:	
Originator:		Technics (R&D)		Origin:	
Target Release:		DIL_OBB_1.0.x		Scope:	
Allocation:		IPS		Verification:	
				See also:	
Issues/Decisions:					
LH Text:					
PH Text:					

3.1.4.3.4 ADM AVAILABILITY

EL4-DIL-SSS-REQ-01481		Availability - ADM - Overall		(Accepted)	
The [ADM] shall have a class 3 overall availability of at least 0,9999.					
Comments:					
Category:		DPR		Priority:	
Originator:		Technics (R&D)		Mandatory	
Target Release:		DIL_OBB_1.0.x		Origin:	
Allocation:		DIL, ADM		Scope:	
				System	
				Verification:	
				< will be added in a later document edition >	
				See also:	
Issues/Decisions:					
LH Text:					
PH Text:					

3.1.4.4 SiDiNET AVAILABILITY

The [SiDiNet] is realised by the customer. The requirements are specified in [22 Network specifications 1] and do not need to be observed by the supplier.

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3.1.4.5 FES AVAILABILITY

EL4-DIL-SSS-REQ-00575 Availability - FES - Overall		(Accepted)
The [FES] shall have a class 2 overall availability of at least 0.99995 (without consideration of the availability of the [OC]s).		
<i>Comments:</i>		
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_124] ==> [EL4 PH] [02_126]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	5.6.0-1 Die Verfügbarkeit des FES muss mindestens 0,99995 betragen. Hierbei muss die Verfügbarkeit der Object Controller nicht in die Betrachtung einbezogen werden. 5.6.0-3 Die FES (Feldelementstation) wird der Verfügbarkeitsklasse 2 zugeordnet.	
<i>PH Text:</i>	Abhängig von der jeweiligen Ausprägung der Feldelementstation und ihrem Einsatzort (Topologie) wird der Schweregrad der betrieblichen Auswirkung betrachtet. Dabei wird berücksichtigt, dass die Konsequenzen eines Ausfalls einer FES nicht von der Anzahl der kontrollierten Elemente alleine (Größe einer FES) sondern auch von ihrem operativen Einsatzort abhängig sind. Die Verfügbarkeitsanforderung wird entsprechend FK1-FK4 festgelegt die den Verfügbarkeitsanforderungen Klasse 2 bis Klasse 5 einer BFZ entsprechen. Die FES (Feldelementstation) wird der Verfügbarkeitsklasse 2 zugeordnet. Hierbei wird die Verfügbarkeit der Object Controller nicht in die Betrachtung einbezogen.	

EL4-DIL-SSS-REQ-00577 Availability distribution of [FES]		(Accepted)
The distribution of the availability of a [FES] shall correspond to class 2. This means in particular an availability of		
0,999979 % (MTTR = 6 h)		
0,99999 % (MTTR = 10 min)		
0,999996 % (MTTR = 2 min)		
<i>Comments:</i>	Equal availabilities with different MTTR values are derived from customer requirements stating that outages with MTTR = 2 min may occur more often per N years than outages with MTTR = 10 min.	
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_127] ==> [EL4 PH] [02_129] ==> [EL4 PH] [02_130]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	5.6.0-4 Die Verfügbarkeit der FES in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 3) muss mindestens 0,999979 betragen. 5.6.0-6 Die Verfügbarkeit der FES in Bezug auf Störungen, welche in der FES innerhalb von 10 Minuten automatisch selbst behoben werden (siehe Tabelle 3), muss mindestens 0,99999 betragen. 5.6.0-7 Die Verfügbarkeit der FES in Bezug auf Störungen, welche in der FES innerhalb von 2 Minuten automatisch selbst behoben werden (siehe Tabelle 3), muss mindestens 0,999996 betragen.	
<i>PH Text:</i>	Die Verfügbarkeit der FES in Bezug auf manuell zu behebbende Störungen muss mindestens 0,999979 betragen. Die Verfügbarkeit der FES in Bezug auf Störungen, welche in der FES innerhalb von 10 Minuten automatisch selbst behoben werden, muss mindestens 0,99999 betragen. Die Verfügbarkeit der FES in Bezug auf Störungen, welche in der FES innerhalb von 2 Minuten automatisch selbst behoben werden, muss mindestens 0,999996 betragen.	

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EL4-DIL-SSS-REQ-00578 Availability - [FES] - Outages with no operational Impact (Accepted)	
Outages of a [FES] with no operational impact shall have an availability of at least 0,997154.	
Comments: Outages of a [FES] with no operational impact correspond to defect class FK4.	
Category: DPR	Priority: Mandatory
Originator: Customer	Origin: ==> [EL4 PH] [02_128]
Target Release: DIL_OBB_1.0.x	Scope: System
Allocation: DIL, DFE	Verification: < will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	5.6.0-5 Die Verfügbarkeit der FES in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben (siehe Tabelle 3), muss mindestens 0,997154 betragen.
PH Text:	Die Verfügbarkeit der FES in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben, muss mindestens 0,997154 betragen.

3.1.4.6 OC AVAILABILITY

EL4-DIL-SSS-REQ-00581 Availability - OC - Overall (Accepted)	
The [OC]s shall have a class 3 overall availability of at least 0.9999.	
Comments: [OC] refers to [LS], [P], [TDS] and [Gen-IO]	
Category: DPR	Priority: Mandatory
Originator: Customer	Origin: ==> [EL4 PH] [02_132]
Target Release: DIL_OBB_1.0.x	Scope: System
Allocation: DCA_OBB_3.1.0	Verification: < will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	5.7.0-1 Die Verfügbarkeit der OC (LS, P, TDS, Gen IO), Blockkonverter und EKSA-Umsetzer muss mindestens 0,9999 betragen.
	5.7.0-3 OC, Blockkonverter und EKSA-Umsetzer werden der Verfügbarkeitsklasse 3 zugeordnet.
PH Text:	Die Verfügbarkeit der OC (LS, P, TDS, Gen IO), Blockkonverter und EKSA-Umsetzer muss mindestens 0,9999 betragen.

EL4-DIL-SSS-REQ-01435 Availability - Block Converters - Overall (Accepted)	
The block converters shall have a class 3 overall availability of at least 0.9999.	
Comments:	
Category: DPR	Priority: Mandatory
Originator: Customer	Origin: ==> [EL4 PH] [02_132]
Target Release: DIL_OBB_1.0.x	Scope: System
Allocation: DIL, DFE	Verification: < will be added in a later document edition >
See also:	
Issues/Decisions:	
LH Text:	5.7.0-1 Die Verfügbarkeit der OC (LS, P, TDS, Gen IO), Blockkonverter und EKSA-Umsetzer muss mindestens 0,9999 betragen.
	5.7.0-3 OC, Blockkonverter und EKSA-Umsetzer werden der Verfügbarkeitsklasse 3 zugeordnet.
PH Text:	Die Verfügbarkeit der OC (LS, P, TDS, Gen IO), Blockkonverter und EKSA-Umsetzer muss mindestens 0,9999 betragen.

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EL4-DIL-SSS-REQ-01436 Availability - LCSI Converters - Overall		(Accepted)
The [LCSI] converters shall have a class 3 overall availability of at least 0.9999.		
<i>Comments:</i>		
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_132]
<i>Target Release:</i>	DIL_OBB_2.0.0	==> [EL4 PH] [02_134]
<i>Allocation:</i>	DIL, DFE	<i>Scope:</i> System
		<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	5.7.0-1 Die Verfügbarkeit der OC (LS, P, TDS, Gen IO), Blockkonverter und EKSA-Umsetzer muss mindestens 0,9999 betragen.	
	5.7.0-3 OC, Blockkonverter und EKSA-Umsetzer werden der Verfügbarkeitsklasse 3 zugeordnet.	
<i>PH Text:</i>	Die Verfügbarkeit der OC (LS, P, TDS, Gen IO), Blockkonverter und EKSA-Umsetzer muss mindestens 0,9999 betragen.	

EL4-DIL-SSS-REQ-00582 Availability distribution of [OC]		(Accepted)
The distribution of the availability of a [OC] must correspond to class 2. This means in particular an availability of		
0,999957 % (MTTR = 6 h)		
0,99999 % (MTTR = 10 min)		
0,999996 % (MTTR = 2 min)		
<i>Comments:</i>	[OC] refers to [LS], [P], [TDS] and [Gen-IO]	
	Equal availabilities with different MTTR values are derived from customer requirements stating that outages with MTTR = 2 min may occur more often per N years than outages with MTTR = 10 min.	
<i>Category:</i>	DPR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_135]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
	DCA_OBB_3.1.0	
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	5.7.0-4 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 4) muss mindestens 0,999957 betragen.	
<i>PH Text:</i>	Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf manuell zu behebbende Störungen muss mindestens 0,999957 betragen.	

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EL4-DIL-SSS-REQ-01437 Availability distribution of [Block]		(Accepted)
The distribution of the availability of a [Block] must correspond to class 2. This means in particular an availability of		
0,999957 % (MTTR = 6 h)		
0,99999 % (MTTR = 10 min)		
0,999996 % (MTTR = 2 min)		
Comments: Equal availabilities with different MTTR values are derived from customer requirements stating that outages with MTTR = 2 min may occur more often per N years than outages with MTTR = 10 min.		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_135]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, DFE	Verification: < will be added in a later document edition >
See also:		
Issues/Decisions:		
LH Text:	5.7.0-4 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 4) muss mindestens 0,999957 betragen.	
PH Text:	Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf manuell zu behebbende Störungen muss mindestens 0,999957 betragen.	

EL4-DIL-SSS-REQ-01438 Availability distribution of [LCSI]		(Accepted)
The distribution of the availability of a [LCSI] must correspond to class 2. This means in particular an availability of		
0,999957 % (MTTR = 6 h)		
0,99999 % (MTTR = 10 min)		
0,999996 % (MTTR = 2 min)		
Comments: Equal availabilities with different MTTR values are derived from customer requirements stating that outages with MTTR = 2 min may occur more often per N years than outages with MTTR = 10 min.		
Category:	DPR	Priority: Mandatory
Originator:	Customer	Origin: ==> [EL4 PH] [02_135]
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, DFE	Verification: < will be added in a later document edition >
See also:		
Issues/Decisions:		
LH Text:	5.7.0-4 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf manuell zu behebbende Störungen (siehe Tabelle 4) muss mindestens 0,999957 betragen.	
PH Text:	Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf manuell zu behebbende Störungen muss mindestens 0,999957 betragen.	

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EL4-DIL-SSS-REQ-00583 Availability - [OC] - Outages with no operational Impact (Accepted)	
Outages of a [OC] with no operational impact shall have an availability of at least 0,997154.	
<i>Comments:</i> [OC] refers to [LS], [P], [TDS] and [Gen-IO] Outages of a [OC] with no operational impact correspond to defect class FK4.	
<i>Category:</i> DPR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DCA_OBB_3.1.0 <i>Allocation:</i> DIL, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_136] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	5.7.0-5 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben (siehe Tabelle 4), muss mindestens 0,997154 betragen.
<i>PH Text:</i>	Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben, muss mindestens 0,997154 betragen.

EL4-DIL-SSS-REQ-01439 Availability - [Block] - Outages with no operational Impact (Accepted)	
Outages of a [Block] with no operational impact shall have an availability of at least 0,997154.	
<i>Comments:</i> Outages of a [Block] with no operational impact correspond to defect class FK4.	
<i>Category:</i> DPR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DIL, DFE <i>Allocation:</i>	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_136] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	5.7.0-5 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben (siehe Tabelle 4), muss mindestens 0,997154 betragen.
<i>PH Text:</i>	Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben, muss mindestens 0,997154 betragen.

EL4-DIL-SSS-REQ-01440 Availability - [LCSI] - Outages with no operational Impact (Accepted)	
Outages of a [LCSI] with no operational impact shall have an availability of at least 0,997154.	
<i>Comments:</i> Outages of a [LCSI] with no operational impact correspond to defect class FK4.	
<i>Category:</i> DPR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x DIL, DFE <i>Allocation:</i>	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_136] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	5.7.0-5 Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben (siehe Tabelle 4), muss mindestens 0,997154 betragen.
<i>PH Text:</i>	Die Verfügbarkeit des OC, Blockkonverter und EKSA-Umsetzer in Bezug auf Fehler, welche keine betrieblichen Auswirkungen haben, muss mindestens 0,997154 betragen.

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EL4-DIL-SSS-REQ-00586		Availability - OC - SW extension and/or update		(In negotiation)
Extensions and/or modifications of the [SW] of the [OC]s, the block converters and the [LCSI] converters shall be carried out during operational breaks, to avoid a restriction of operation below the permitted availability.				
Comments:		As the [OC] is a non-redundant subsystem any [SW] extension or modification means an interrupt.		
== OPEN POINT ==				
Requirement is different to LH expectation and needs to be refined.				
Category:	DPR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_139]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	5.7.0-8 Erweiterungen und/oder Änderungen der Software von OC, Blockkonverter und EKSA-Umsetzer (Software Updates/Upgrades, Projektierungs-, oder Konfigurationsänderungen) dürfen nicht zur Einschränkung des Betriebs unter die erlaubte Verfügbarkeit führen.			
PH Text:	Die Einschränkung des Betriebs und Unterschreitung unter die erlaubte Verfügbarkeit ist in Abhängigkeit zur Anzahl und des Auswirkungsgrades der notwendigen Erweiterungen und Änderungen der OC-Software zu bestimmen. Gegebenenfalls sind entsprechende Wartungsfenster außerhalb der Betriebszeiten zu definieren.			

3.1.4.7 LRU AVAILABILITY

EL4-DIL-SSS-REQ-00313		Availability - LRU - No Maintenance Impact		(Accepted)	
A corrective maintenance action of a [LRU] shall not affect the operational availability of other [LRU]s.					
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).			
Category:	DPR	Priority:	Mandatory		
Originator:	RAM	Origin:			
Target Release:	DIL_OBB_1.0.x	Scope:	System		
Allocation:	DIL, DFE	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:					
PH Text:					

3.1.4.8 MONITORING SYSTEM [FRACAS]

The customer provides the supplier with the necessary data from the [MDM]/[DDSP] for the availability verification.

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EL4-DIL-SSS-REQ-00671		Availability - FRACAS - Compliance with Standards		(Accepted)			
The supplier shall provide the customer with a [FRACAS] system in accordance with [CENELEC] standard [EN50126-1] for all systems and subsystems to be delivered.							
Comments:							
Category:		MDR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_141]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		5.8.0-1 Der AN muss ein FRACAS-System gemäß CENELEC-Norm EN50126-1 dem AG für alle zu liefernden Systeme und Subsysteme zur Verfügung stellen.					
PH Text:		Über ein vom AN eingerichtetes FRACAS-System werden alle Hardware- und Softwarefehler der Kategorie FK1-FK4 während des Betriebs ausgewertet. Es werden alle ergriffenen Korrekturmaßnahmen aufgezeichnet und die so ermittelten Ausfalldaten vom RAM-Management für regelmäßige Auswertung herangezogen.					

EL4-DIL-SSS-REQ-00672		Availability - FRACAS - Proof		(Accepted)			
The supplier shall prove by means of [FRACAS] that the required availability values ([DC], [VIL], [FES], [OC]) have been met within 5 years of the first [VIL] being put into operation.							
Comments:							
Category:		MDR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_142]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL, VIL, IPS, ADM, DFE		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		5.8.0-2 Der AN muss mittels FRACAS die Einhaltung der geforderten Verfügbarkeitswerte (Rechenzentrum, VSTW, FES, OC) innerhalb von 5 Jahren nach IBN des ersten VSTW nachweisen.					
PH Text:		Eine Einhaltung der vom AG geforderten Verfügbarkeitswerte (Rechenzentrum, VSTW, FES, OC) wird innerhalb von 5 Jahren nach IBN des ersten VSTW vom AG unter Berücksichtigung folgender Parameter nachgewiesen: - Beobachtungsdauer, - Anzahl der ausgefallenen Einheiten innerhalb der Beobachtungsdauer, - Anzahl der Einheiten zu Beginn des Beobachtungszeitraums.					

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EL4-DIL-SSS-REQ-00673		Availability - FRACAS - Monitoring and Verifying		(Accepted)	
The [FRACAS] system shall be used to monitor the availability of each individual [DIL] components and to verify compliance with the availability requirements.					
Comments:					
Category: MDR		Priority: Mandatory			
Originator: Customer		Origin: ==> [EL4 PH] [02_143]			
		==> [EL4 PH] [02_147]			
Target Release: DIL_OBB_1.0.x		Scope: System			
Allocation: DIL, VIL, IPS, ADM, DFE		Verification: < will be added in a later document edition >			
		See also:			
Issues/Decisions:					
LH Text:		5.8.0-3 Mittels dem FRACAS-System muss die Verfügbarkeit jeder einzelnen DSTW-Komponenten (VSTW, Feldelementstation, OC) überwacht und die Einhaltung der Verfügbarkeitsanforderungen nachgewiesen werden.			
		5.8.0-7 Der AN muss die Verfügbarkeit der einzelnen, gelieferten Systeme überwachen und nachweisen.			
PH Text:		Die über ein vom AN eingerichtetes FRACAS-System gewonnenen empirischen Ausfallsraten einzelner DSTW-Komponenten dienen zum Nachweis der Einhaltung der vom AG geforderten Verfügbarkeitsanforderungen (VSTW, Feldelementstation, OC).			
		Jedes einzelne der gelieferten Systeme wird diagnostisch im Hinblick auf seine Verfügbarkeit überwacht.			
		Die über ein vom AN eingerichtetes FRACAS-System gewonnenen empirischen Ausfallsraten einzelner DSTW-Komponenten dienen zum Nachweis der Einhaltung der vom AG geforderten Verfügbarkeitsanforderungen.			

EL4-DIL-SSS-REQ-00674		Availability - FRACAS - Regular Reports		(Accepted)			
The [FRACAS] system shall provide the customer with regular reports.							
Comments:							
Category:		MDR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_144]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL, VIL, IPS, ADM, DFE		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		5.8.0-4 Das FRACAS-System muss dem AG regelmäßige Reports bereitstellen.					
PH Text:		Basierend auf dem vom AN eingerichteten FRACAS-System werden dem AG entsprechend der zeitlichen Vorgaben regelmäßige Reports präsentiert und zur Verfügung gestellt.					

EL4-DIL-SSS-REQ-00675		Availability - FRACAS - Display of Information		(Accepted)			
In reporting, for example, information such as the availability of classes 1 to 3 shall be displayed. In doing so, the information on availability down to the module level (module -> see definition [DIL]) shall be displayed/shown with the information on the reason for [failure].							
Comments:							
Category:		MDR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_145]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL, VIL, IPS, ADM, DFE		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		5.8.0-5 Im Berichtswesen sollen beispielsweise Angaben wie Verfügbarkeit der Klassen 1 bis 3 angezeigt werden. Dabei sollte weitestgehend die Angabe der Verfügbarkeit bis in die Modulebene (Modul siehe Definition DSTW) mit der Angabe des Ausfallgrundes angezeigt/ausgewiesen werden.					
PH Text:		Der AN wird ein Berichtswesen in dieser Art umsetzen.					

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EL4-DIL-SSS-REQ-00676		Availability - FRACAS - SLAs		(In negotiation)
The required availability shall be maintained in line with the availability of OBB's standard [fault] clearance processes: → [SLA] 5 minutes for [VIL] → [SLA] 30 minutes for [FES] and [OC].				
<i>Comments:</i> == OPEN POINT == Has to clarified with the customer. E.g. When does the time frame starts? From the arrival or from the moment the [fault] appears?				
<i>Category:</i> MDR		<i>Priority:</i> Mandatory		
<i>Originator:</i> Customer		<i>Origin:</i> ==> [EL4 PH] [02_146]		
<i>Target Release:</i> DIL_OBB_1.0.x		<i>Scope:</i> System		
<i>Allocation:</i> DIL, VIL, IPS, ADM, DFE		<i>Verification:</i> < will be added in a later document edition >		
<i>See also:</i>				
<i>Issues/Decisions:</i>				
<i>LH Text:</i>		5.8.0-6 Die geforderte Verfügbarkeit ist entsprechend der Verfügbarkeit der Standard ÖBB-Entstörprozesse einzuhalten: • SLA 5 Minuten: VSTW; • SLA 30 Minuten: Feldelementstation und OC.		
<i>PH Text:</i>		siehe Kommentar.		

EL4-DIL-SSS-REQ-00678		Availability - FRACAS - Monitoring, Documentation and Analysis		(Accepted)
The system integrator (supplier of the [DIL]) shall monitor, document and analyse the availability of the overall system, including all subsystems, and communicate the results to the customer.				
Comments:				
Category:	MDR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_149]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, VIL, IPS, ADM, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	5.8.0-9 Der Systemintegrator (Lieferant des VSTW) muss die Verfügbarkeit des Gesamtsystems inklusive aller Teilsysteme überwachen, dokumentieren, analysieren und die Ergebnisse dem AG übermitteln.			
PH Text:	Jedes einzelne der gelieferten Systeme wird diagnostisch im Hinblick auf seine Verfügbarkeit überwacht, dokumentiert, analysiert und die Ergebnisse werden dem AG übermittelt. Die über ein vom AN eingerichtetes FRACAS-System gewonnenen empirischen Ausfallsraten einzelner DSTW-Komponenten dienen zum Nachweis der Einhaltung der vom AG geforderten Verfügbarkeitsanforderungen.			

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EL4-DIL-SSS-REQ-00679		Availability - FRACAS - Coordination due to Non-Compliance		(Accepted)	
The system integrator (supplier of the [VIL]) shall be responsible for any necessary coordination due to non-compliance with availability.					
Comments:					
Category:		MDR		Priority:	
Originator:		Customer		Mandatory	
Target Release:		DIL_OBB_1.0.x		Origin:	
Allocation:		DIL, VIL, IPS, ADM, DFE		==> [EL4 PH] [02_150]	
				Scope:	
				System	
				Verification:	
				< will be added in a later document edition >	
				See also:	
Issues/Decisions:					
LH Text:		5.8.0-10 Die Verantwortung für jegliche notwendige Abstimmung, Aufgrund von nicht einhalten der Verfügbarkeit, liegt beim Systemintegrator (Lieferant des VSTW). Der Lieferant der Teilsysteme hat konstruktive Mitwirkungspflicht bei diesen Abstimmungen.			
PH Text:		siehe Kommentar			

EL4-DIL-SSS-REQ-00680		Availability - FRACAS - Cooperation due to Non-Compliance		(Accepted)	
The supplier of the subsystems shall have a constructive obligation to cooperate in the coordination measures mentioned in [EL4-DIL-SSS-REQ-00679].					
Comments:					
Category:		MDR		Priority:	
Originator:		Customer		Mandatory	
Target Release:		DIL_OBB_1.0.x		Origin:	
Allocation:		DIL, VIL, IPS, ADM, DFE		==> [EL4 PH] [02_150]	
				Scope:	
				System	
				Verification:	
				< will be added in a later document edition >	
				See also:	
Issues/Decisions:					
LH Text:		5.8.0-10 Die Verantwortung für jegliche notwendige Abstimmung, Aufgrund von nicht einhalten der Verfügbarkeit, liegt beim Systemintegrator (Lieferant des VSTW). Der Lieferant der Teilsysteme hat konstruktive Mitwirkungspflicht bei diesen Abstimmungen.			
PH Text:		siehe Kommentar			

EL4-DIL-SSS-REQ-00681		Availability - FRACAS - Responsibility		(Accepted)			
The supplier shall have the responsibility for the availability of the system delivered.							
Comments:							
Category:		MDR		Priority:		Mandatory	
Originator:		Customer		Origin:		==> [EL4 PH] [02_257]	
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		DIL, VIL, IPS, ADM, DFE		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:		5.8.0-11 Die Verantwortung für die Verfügbarkeit des vom AN gelieferten Systems liegt beim AN.					
PH Text:		siehe Kommentar					

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3.1.4.9 POWER SUPPLY

EL4-DIL-SSS-REQ-00308 Power supply - UPS Design (Accepted)	
The three phases of each [UPS] shall be connected with the corresponding PDU in a way to tolerate HW (failures) if possible.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> Product Management <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, IPS, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

EL4-DIL-SSS-REQ-00422 Power supply - PDU Design (Accepted)	
Each PDU shall be protected separately by means of using a circuit breaker.	
<i>Comments:</i> • This means that an outage of one power supply phase has no impact on other phases. • This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, VIL, IPS, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

EL4-DIL-SSS-REQ-00309 Power supply - Availability related Components - Redundancy (Accepted)	
All critical HW components which are related to availability (OCs, switches), shall be equipped with two independent power supplies.	
<i>Comments:</i> This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).	
<i>Category:</i> DPR <i>Originator:</i> Product Management <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, IPS, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> System <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
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EL4-DIL-SSS-REQ-00315 Power Supply - Non-Availability related Components - Monitoring and Backup		(Accepted)
Components not related to availability shall do with one power supply, but in that case monitoring and backup equipment for automatic switch-over shall be provided.		
Comments: This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category:	DPR	Priority: Mandatory
Originator:	RAM	Origin:
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, IPS, DFE	Verification: < will be added in a later document edition >
		See also:
Issues/Decisions:		
LH Text:		
PH Text:		

3.1.4.10 NETWORK AND SWITCHES

EL4-DIL-SSS-REQ-00307 Availability - Network - Overall		(Accepted)
The network availability of a [DIL] location shall be at least 0,999999.		
Comments: This requirement refers to the internal network, i.e., DIL-CU. The network availability was deliberately set 10 times higher than the most stringent availability requirement for the entire system (Class 1 = 0.99999) to ensure overall system availability. This assumption is based on Nokia network availability calculations (analogous to OMC). This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category:	DPR	Priority: Mandatory
Originator:	RAM	Origin:
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, IPS	Verification: < will be added in a later document edition >
		See also:
Issues/Decisions:		
LH Text:		
PH Text:		

EL4-DIL-SSS-REQ-00310 Availability - Switches - Connection with Servers		(Accepted)
A server within a server cabinet shall be connected to each switch within this server cabinet.		
Comments: This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
Category:	GCR	Priority: Mandatory
Originator:	Product Management	Origin:
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	DIL, IPS	Verification: < will be added in a later document edition >
		See also:
Issues/Decisions:		
LH Text:		
PH Text:		

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EL4-DIL-SSS-REQ-00311		Availability - Switches - Internal Network - Connection with Routers		(Accepted)	
The switches for the internal network within a cabinet shall be connected with the two routers of the network cabinet.					
Comments:		This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).			
Category:	GCR	Priority:	Mandatory		
Originator:	Product Management	Origin:			
Target Release:	DIL_OBB_1.0.x	Scope:	System		
Allocation:	DIL, IPS	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:					
PH Text:					

3.1.5 RAM DESIGN RULES

The [RAM] Design Rules and the necessary definitions to interpret the rules correctly are stated in the project specific [RAM] Plan and also in the corresponding [SSDD] documents.

These are rules for:

- Balanced distribution
- Balanced increment
- Affinity
- Anti-Affinity
- Equi-Distribution
- Topology Spread

== OPEN POINT ==

Update of the chapter

EL4-DIL-SSS-REQ-01483		Server distribution		(Accepted)			
The placement of the servers of the same type across the availability zones shall follow the balanced distribution principle. Server types are control server, compute server and storage server							
Comments:							
Category:		GCR		Priority:		Mandatory	
Originator:		RAM		Origin:			
Target Release:		DIL_OBB_1.0.x		Scope:		System	
Allocation:		IPS		Verification:		< will be added in a later document edition >	
				See also:			
Issues/Decisions:							
LH Text:							
PH Text:							

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EL4-DIL-SSS-REQ-01484 Server reintegration (Accepted)	
A reintegration of one or more servers of the same type shall follow the balanced increment principle.	
<i>Comments:</i>	
<i>Category:</i> GCR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> IPS	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

EL4-DIL-SSS-REQ-01485 Server failure (Accepted)	
In case of a server failure the services from this server shall be redistributed to the remaining servers following the balanced increment principle. This ensures a balanced distribution of the services on the remaining servers.	
<i>Comments:</i>	
<i>Category:</i> GCR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> IPS	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

EL4-DIL-SSS-REQ-01486 No service redistribution after server reintegration (Accepted)	
In case of reintegration of a server no services shall be redistributed. This prevents any additional disruption of operating services.	
<i>Comments:</i>	
<i>Category:</i> GCR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> IPS	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

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EL4-DIL-SSS-REQ-01487 Mapping component failure to severity level (Accepted)	
The operative consequences of a failed hardware component shall be defined by the severity level of the availability class.	
<i>Comments:</i>	
<i>Category:</i> GCR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> IPS	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

EL4-DIL-SSS-REQ-01488 Mapping corrective maintenance action to severity level (Accepted)	
The operative consequences during a corrective maintenance activity shall be defined by the severity level of the availability class.	
<i>Comments:</i>	
<i>Category:</i> GCR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> IPS	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

EL4-DIL-SSS-REQ-01489 Enable mapping of severity levels (Accepted)	
The architectural concept shall be defined in a way to enable a mapping of severity levels to failed / reintegrated components.	
<i>Comments:</i> For instance redundancy properties like 1oo2, etc. and properties like Anti-Affinity, Equi-Distribution, etc. should be clearly stated in architecture documents for referencing.	
<i>Category:</i> GCR <i>Originator:</i> RAM <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> IPS	<i>Priority:</i> Mandatory <i>Origin:</i> <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	
<i>PH Text:</i>	

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EL4-DIL-SSS-REQ-01490 CG - balanced distribution		(Accepted)
CGs representing <VILs> shall be placed on different physical servers and on different availability zones following the balanced distribution principle.		
Comments: Equi-Distribution of CGs on physical servers and Topology spread constraint over availability zones		
Category:	GCR	Priority: Mandatory
Originator:	RAM	Origin:
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	IPS	Verification: < will be added in a later document edition >
See also:		
Issues/Decisions:		
LH Text:		
PH Text:		

EL4-DIL-SSS-REQ-01491 CN - balanced distribution		(Accepted)
CNs belonging to the same CG shall be placed on different physical servers and on different availability zones following the balanced distribution principle.		
Comments: Equi-Distribution of CNs on physical servers and Topology spread constraint over availability zones and Anti-Affinity CN		
Category:	GCR	Priority: Mandatory
Originator:	RAM	Origin:
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	IPS	Verification: < will be added in a later document edition >
See also:		
Issues/Decisions:		
LH Text:		
PH Text:		

EL4-DIL-SSS-REQ-01492 CE - balanced distribution		(Accepted)
CEs belonging to a CN shall be placed on different physical servers at the same availability zone following the balanced distribution principle.		
Comments: Equi-Distribution of CEs on physical servers and Anti-Affinity CE		
Category:	GCR	Priority: Mandatory
Originator:	RAM	Origin:
Target Release:	DIL_OBB_1.0.x	Scope: System
Allocation:	IPS	Verification: < will be added in a later document edition >
See also:		
Issues/Decisions:		
LH Text:		
PH Text:		

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EL4-DIL-SSS-REQ-01482 [VIL] Applications		(Accepted)
The VIL-applications which belong to a specific VIL-instance should be executed on the same physical server		
<i>Comments:</i>		
<i>Category:</i>	GCR	<i>Priority:</i> Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i>
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	VIL, IPS	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>		
<i>PH Text:</i>		

3.1.6 SAFETY

3.1.6.1 SAFETY REQUIREMENTS [DIL]

The following table contains safety-relevant [DIL] functions and subfunctions or safety requirements identified in the context of [EULYNX].

Threshold values for availability [THR] were determined for each individual subsystem and for each corresponding interface based on the safety requirements of the [DIL].

In accordance with [OEVE/OENORM EN 50129], each safety-related subsystem and the interfaces were assigned a **safety integrity level [SIL]**.

For details concerning the consideration of the safety integrity level refer to the corresponding chapters of the [DIL] system and subsystem requirement specifications.

Table 4 Safety Requirements

ID	Safety Relevant Function	DIL-CU	Maintenance and Diagnosis	Light Signal	TD\$	Point	I/O	Pos	SCWS (X25oRa\$TA)	RBC (X25oRa\$TA)	LCSi (SCL-LX)	TCs (X25oRa\$TA)	NIL (X25oRa\$TA/X25-Block)	CDC (EIL 2)	SCL-P	SCL-O	SCL-S	SCL-TD\$	p3	IO2	IO3	LS2	TD\$2	SCLACS
B.2	Protecting the route	SIL 4	-	SIL 3	SIL 4	SIL 4	SIL 4	SIL 4	-	SIL 4	-	BI	SIL 4	-	SIL 4	SIL 4	SIL 3	SIL 4	SIL 4	SIL 4	SIL 4	SIL 4	SIL 4	SIL 4
B.3	Protection at same-level crossings	SIL 4	-	-	-	-	SIL 4	SIL 4	-	-	SIL 4	-	-	-	-	SIL 4	-	-	-	SIL 4	SIL 4	-	-	-
B.4	Ensuring the distance keeping	SIL 4	-	SIL 4	SIL 4	-	-	SIL 4	-	SIL 4	-	-	SIL 4	-	-	-	SIL 4	SIL 4	-	-	-	SIL 4	SIL 4	SIL 4
B.5	Supervision of the speed	SIL 4	-	SIL 4	-	-	-	SIL 4	-	SIL 4	-	-	SIL 4	-	-	-	SIL 4	-	-	-	-	SIL 4	-	-
B.6	Protection against obstacles in the route	SIL 4	-	-	-	-	-	SIL 4	-	-	SIL 3	SIL 4	-	-	-	-	-	-	-	-	-	-	-	-
B.7	Operation and display	SIL 4	-	-	-	-	-	SIL 4	-	-	-	SIL 4	-	-	-	-	-	-	-	-	-	-	-	-
B.8	Automisation of operation	BI	-	-	-	-	-	BI	-	-	-	BI	-	-	-	-	-	-	-	-	-	-	-	-
B.9	Supervision of the operation	BI	-	-	-	-	-	BI	-	-	-	BI	-	-	-	-	-	-	-	-	-	-	-	-

Part systems
Surrounding systems
Internal ports
External ports

Figure 2 Satey Relevant Functions and SIL

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3.1.6.2 TOLERABLE HAZARD RATES

EL4-DIL-SSS-REQ-00588 Safety - THR - VIL		(Accepted)
The [THR] value of the subsystem [VIL] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_158]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, VIL	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	6.2.0-1 Der THR Wert des Teilsystems VSTW muss kleiner gleich 1.0E-9 1/h sein	
<i>PH Text:</i>	Das geforderte Sicherheitsziel wird für die Komponente eingehalten.	

EL4-DIL-SSS-REQ-00589 Safety - THR - LS		(Accepted)
The [THR] value of the subsystem [LS] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_160]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DCA_OBB_3.1.0 DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	6.3.0-1 Der THR Wert des Teilsystems LS muss kleiner gleich 1.0E-9 1/h sein	
<i>PH Text:</i>	Das geforderte Sicherheitsziel wird für die Komponente eingehalten.	

EL4-DIL-SSS-REQ-00590 Safety - THR - P		(Accepted)
The [THR] value of the subsystem [P] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_162]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DCA_OBB_3.1.0 DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
<i>See also:</i>		
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	6.4.0-1 Der THR Wert des Teilsystems Weiche muss kleiner gleich 1.0E-9 1/h sein	
<i>PH Text:</i>	Das geforderte Sicherheitsziel wird für die Komponente eingehalten.	

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EL4-DIL-SSS-REQ-00591 Safety - THR - TDS		(Accepted)
The [THR] value of the subsystem [TDS] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_164]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	6.5.0-1 Der THR Wert des Teilsystems TDS muss kleiner gleich 1.0E-9 1/h sein	
<i>PH Text:</i>	Das geforderte Sicherheitsziel wird für die Komponente eingehalten.	

EL4-DIL-SSS-REQ-00592 Safety - THR - IO		(Accepted)
The [THR] value of the subsystem [IO] shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_166]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
	DCA_OBB_3.1.0	
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	6.6.0-1 Der THR Wert des Teilsystems EA muss kleiner gleich 1.0E-9 1/h sein	
<i>PH Text:</i>	Das geforderte Sicherheitsziel wird für die Komponente eingehalten.	

EL4-DIL-SSS-REQ-01495 Safety - THR - IO		(Accepted)
The [THR] value of the subsystem block converter shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>		
<i>PH Text:</i>		

EL4-DIL-SSS-REQ-00593 Safety - THR - LCSi		(Accepted)
The [THR] value of the [LCSi] Converter shall be less or equal 1.0E-9 1/h.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_168]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	6.7.0-1 Der THR Wert des EKSA-Umsetzers muss kleiner gleich 1.0E-9 1/h sein	
<i>PH Text:</i>	Das geforderte Sicherheitsziel wird für die Komponente eingehalten.	

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3.1.6.3 INTERFACE - INTEGRITY LEVELS

EL4-DIL-SSS-REQ-00735 Safety - Integrity Levels - Interface - X25oRaSTA, (Accepted) EULYNX - Functional Hazard Rates			
The use of the required interfaces ([X25oRaSTA], [EULYNX]) shall be carried out in such a way that the functional hazard rates of the involved subsystems are not increased.			
<i>Comments:</i> == MISMATCH == Das ist eine Anforderung, aber im LH nur als "Info" markiert.			
<i>Category:</i>	FUR	<i>Priority:</i>	Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_170]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i>	System
<i>Allocation:</i>	DIL, VIL, DFE	<i>Verification:</i>	< will be added in a later document edition >
		<i>See also:</i>	
<i>Issues/Decisions:</i>			
<i>LH Text:</i>	6.8.0-1 Der Einsatz der erforderlichen Schnittstellen (X25oRaSTA, EULYNX) muss derart erfolgen, dass die funktionalen Gefährdungsraten der beteiligten Subsysteme nicht erhöht werden.		
<i>PH Text:</i>			

EL4-DIL-SSS-REQ-00683 Safety - Integrity Levels - Interface - Adjacent (Accepted) Systems			
The interfaces of the [VIL] to the adjacent systems [ILS], [RBC], [LCSI], [SCWS] (Trackworker Safety System) and [OMC] ([TCS]) shall meet the requirements of Safety Integrity Level 4 [SIL]4.			
Comments:			
Category:	SAR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_171]
Target Release:	DIL_OBB_1.0.x	Scope:	System
Allocation:	DIL, VIL, DFE	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	6.8.0-2 Die Schnittstellen des VSTW zu den Umsystemen ILS, RBC, EKSA, SCWS (Trackworker Safety System) und BFZ (TCS) müssen Anforderungen gemäß dem Safety Integrity Level 4 (SIL4) erfüllen.		
PH Text:	Die Schnittstellen des VSTW zu den Umsystemen ILS, RBC, EKSA, SCWS (Trackworker Safety System) und BFZ (TCS) werden die Anforderungen gemäß dem Safety Integrity Level 4 (SIL4) erfüllen.		

EL4-DIL-SSS-REQ-00684 Safety - Integrity Levels - Interface - OC (Accepted)			
The interfaces of the [VIL] [SCI-LS], [SCI-TDS], [SCI-P], [SCI-IO] to the [OC]s light signal, track detection system, points, generic IO shall fulfil the requirements according to Safety Integrity Level 4 [SIL]4.			
Comments:			
Category:	SAR	Priority:	Mandatory
Originator:	Customer	Origin:	==> [EL4 PH] [02_172]
Target Release:	DIL_OBB_1.0.x	Scope:	System
Allocation:	DCA_OBB_3.1.0 DIL, VIL, DFE	Verification:	< will be added in a later document edition >
See also:			
Issues/Decisions:			
LH Text:	6.8.0-3 Die Schnittstellen des VSTW SCI-LS, SCI-TDS, SCI-P, SCI-IO zu den OCs Lichtsignal, Gleisfreimeldung, Weiche, Ein-/und Ausgabeelement müssen Anforderungen gemäß dem Safety Integrity Level 4 (SIL4) erfüllen.		
PH Text:	Die Schnittstellen des VSTW SCI-LS, SCI-TDS, SCI-P, SCI-IO zu den OCs Lichtsignal, Gleisfreimeldung, Weiche, Ein-/und Ausgabeelement werden die Anforderungen gemäß dem Safety Integrity Level 4 (SIL4) erfüllen.		

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EL4-DIL-SSS-REQ-00685		Safety - Integrity Levels - Interface - DDSP, PS		(Accepted)
The [VIL] - [DDSP] and [VIL] - [PS] interfaces shall meet the requirements for basic integrity [BI].				
Comments:				
Category:	SAR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_174]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, IPS	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	6.9.0-1 Die Schnittstellen VSTW - DDSP und VSTW - Stromversorgung müssen Anforderungen gemäß der Basis Integrität (BI) erfüllen.			
PH Text:	Die Schnittstellen VSTW - DDSP und VSTW - Stromversorgung werden gemäß den Anforderungen nach Basis Integrität (BI) entwickelt.			

EL4-DIL-SSS-REQ-00686		Safety - Integrity Levels - Development DIL		(Accepted)	
The development of the [DIL] shall be carried out in accordance with the safety requirement level [SIL]4 and the procedures and techniques requested in [OEVE/OENORM EN 50129] shall be taken into account.					
Comments:		Compliance with the requirements of [OEVE/OENORM EN 50129] is documented in the Safety Plan.			
Category:	SAR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_175]		
Target Release:	DIL_OBB_1.0.x	Scope:	System		
	DCA_OBB_3.1.0				
Allocation:	DIL, VIL, DFE	Verification:	< will be added in a later document edition >		
		See also:	==> [EL4-DIL-SSS-REQ-00305] Fulfillment of Standard EN 50129		
Issues/Decisions:					
LH Text:	6.9.0-2 Die Entwicklung des DSTW muss gemäß der Sicherheitsanforderungsstufe SIL4 erfolgen und die dafür in [OEVE/OENORM EN 50129] vorgeschriebenen Verfahren und Techniken berücksichtigen.				
PH Text:	Das DSTW wird gemäß [OEVE/OENORM EN 50129] entwickelt. Details zur Erfüllung der Anforderungen aus [OEVE/OENORM EN 50129] werden in einem Safety Plan dargelegt.				

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EL4-DIL-SSS-REQ-00687 Safety - Integrity Levels - SW VIL, OC		(Accepted)
The software of the [VIL] and the [OC]s shall be developed in accordance with the specified software safety requirements and the procedures and techniques requested in [OEVE/OENORM EN 50716] for it.		
Comments: OEVE EN 50716 was published on October 1, 2024, specifying the withdrawal of EN 50128 at the end of October 2026. Therefore, EN 50716 shall be used for the DIL. Compliance with the requirements of [OEVE/OENORM EN 50716] is documented in the Safety Plan.		
Category: SAR Originator: Customer Target Release: DIL_OBB_1.0.x DCA_OBB_3.1.0 Allocation: DIL, VIL, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_176] Scope: System Verification: < will be added in a later document edition > See also: ==> [EL4-DIL-SSS-REQ-00304] Fulfillment of Standard EN 50716	
Issues/Decisions:		
LH Text:	6.9.0-3 Die Software-Entwicklung des VSTW und der OCs muss gemäß der festgelegten Software-Sicherheitsanforderungsstufe erfolgen und die dafür in [OEVE/OENORM EN 50128] vorgeschriebenen Verfahren und Techniken berücksichtigen.	
PH Text:	Sicherheitsrelevante Software wird gemäß [OEVE/OENORM EN 50716] entwickelt. Details zur Erfüllung der Anforderungen aus [OEVE/OENORM EN 50716] werden in einem Safety Plan dargelegt.	

EL4-DIL-SSS-REQ-00688 Safety - Integrity Levels - Certificates		(Accepted)
The supplier/manufacture of the [DIL] shall prepare the safety certificates in accordance with [OEVE/OENORM EN 50129].		
Comments: Compliance with the requirements of [OEVE/OENORM EN 50129] is documented in the Safety Plan.		
Category: SAR Originator: Customer Target Release: DIL_OBB_1.0.x DCA_OBB_3.1.0 Allocation: DIL, DFE	Priority: Mandatory Origin: ==> [EL4 PH] [02_177] Scope: System Verification: < will be added in a later document edition > See also:	
Issues/Decisions:		
LH Text:	6.9.0-4 Der Lieferant/Hersteller des DSTW muss die Sicherheitsnachweise gemäß [OEVE/OENORM EN 50129] erstellen.	
PH Text:	Für das DSTW wird ein Sicherheitsnachweis gemäß [OEVE/OENORM EN 50129] erstellt. Details dazu werden in einem Safety Plan dargelegt.	

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EL4-DIL-SSS-REQ-00689		Safety - Integrity Levels - Data Communication		(In negotiation)
For data communication via open (categories 2 and 3) and closed (category 1) networks, [OEVE/OENORM EN 50159] shall be complied with. The OCs shall provide mechanism for secure data communication over category 3 networks.				
<i>Comments:</i> == OPEN POINT == Definition of Classes for Network required. SiDi Net Class 3 DN2020? X25oRaSTA Class 2				
<i>Category:</i>	SAR	<i>Priority:</i>	Mandatory	
<i>Originator:</i>	Customer	<i>Origin:</i>	==> [EL4 PH] [02_178]	
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i>	System	
<i>Allocation:</i>	DCA_OBB_3.1.0 DIL, VIL, DFE	<i>Verification:</i>	< will be added in a later document edition > <i>See also:</i>	
<i>Issues/Decisions:</i>				
<i>LH Text:</i>	6.9.0-5 Für Datenkommunikation über offene und geschlossene Netze muss [OEVE/OENORM EN 50159] eingehalten werden.			
<i>PH Text:</i>	Die Datenkommunikation wird die Anforderungen von [OEVE/OENORM EN 50159] erfüllen.			

EL4-DIL-SSS-REQ-00690		Safety - Integrity Levels - Misuse of Track Elements		(Accepted)
The supplier/manufacturer of the [DIL] shall agree and define measures with the customer to protect against misuse of the trackside elements.				
Comments:				
Category:	SAR	Priority:	Mandatory	
Originator:	Customer	Origin:	==> [EL4 PH] [02_179]	
Target Release:	DIL_OBB_1.0.x	Scope:	System	
Allocation:	DIL, DFE	Verification:	< will be added in a later document edition >	
		See also:		
Issues/Decisions:				
LH Text:	6.9.0-6 Der Lieferant/Hersteller des DSTW muss Maßnahmen zum Schutz vor Missbrauch der streckenseitigen Elemente mit dem AG abstimmen und festlegen.			
PH Text:	Mögliche Maßnahmen aus Sicht des AN: -Versperrbare bzw. zutrittskontrollierte FES in Abhängigkeit der Größe -Aktive Überwachung der Interfaces -Verwendung proprietärer Interfaces oder Protokolle in der Feldelementansteuerung bzw. Kommunikaton			

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3.1.6.4 MAINTENANCE

EL4-DIL-SSS-REQ-00691 Safety - Maintenance - SRACs (Accepted)	
Any additional expenses, efforts or [SRAC]s for the maintenance personnel of the customer as a result of the [CENELEC] phases 5-10 shall be agreed between the supplier and the customer.	
<i>Comments:</i>	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_181] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	6.10.0-1 Für aus den CENELEC-Phasen 5-10 resultierende zusätzlichen Aufwendungen oder SRACs an das Instandhaltungs-Personal des AG müssen zwischen AN und AG abgestimmt werden.
<i>PH Text:</i>	Hitachi Rail hat das Ziel, keine SRACs an die Instandhaltung weiterzugeben und insbesondere nicht die Verantwortung von technisch lösbaren Problemen durch ein SRAC an die Instandhaltung abzudecken. Falls die Notwendigkeit für ein SRAC an die Instandhaltung (bzw. generell an den AG) besteht, wird dieses immer nur in Abstimmung mit dem AG erfolgen.

EL4-DIL-SSS-REQ-00692 Safety - Maintenance - Activities (Accepted)	
It shall not be possible for the planned maintenance activities to result in unsafe conditions in terms of safety or operation.	
<i>Comments:</i>	
<i>Category:</i> SAR <i>Originator:</i> Customer <i>Target Release:</i> DIL_OBB_1.0.x <i>Allocation:</i> DIL, DFE	<i>Priority:</i> Mandatory <i>Origin:</i> ==> [EL4 PH] [02_182] <i>Scope:</i> System <i>Verification:</i> < will be added in a later document edition > <i>See also:</i>
<i>Issues/Decisions:</i>	
<i>LH Text:</i>	6.10.0-2 Durch die Durchführung von geplanten Instandhaltungstätigkeiten dürfen keine sicherungstechnisch und betrieblich unsicheren Zustände hergestellt werden können.
<i>PH Text:</i>	Unter der Voraussetzung der konformen Anwendung der Wartungs- und Instandhaltungsdokumentation kann bei der Durchführung von geplanten Instandhaltungstätigkeiten kein unsicherer Zustand hergestellt werden.

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EL4-DIL-SSS-REQ-00693		Safety - Maintenance - Exchange of FES elements		(Accepted)	
The exchange of systems, subsystems or components of the customer in [FES] or server racks of the supplier shall not require any safety-related training or education.					
Comments:		The customer does not consider [DIL] (basic) training according to the supplier's concept as safety-related training or education.			
Category:	SAR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_253] ==> [EL4 PH] [02_252]		
Target Release:	DIL_OBB_1.0.x	Scope:	System		
Allocation:	DIL, DFE	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:		6.10.0-3 Eine DSTW-(Basis)Schulung laut Konzept des AN ist nicht als sicherungstechnische Schulung oder Ausbildung des AG anzusehen. 6.10.0-4 Tausch von Systemen, Teilsystemen oder Komponenten des AG in FES oder Server-Racks des AN dürfen keine sicherungstechnische Schulung beziehungsweise Ausbildung erfordern.			
PH Text:		Die Teilsysteme oder Komponenten des FES und der Serveranlage werden so entwickelt, dass keine sicherungstechnische Schulung beziehungsweise Ausbildung für den Tausch erforderlich sein wird.			

3.1.6.5 MISCELLANEOUS

EL4-DIL-SSS-REQ-00694		Safety - Power Supply - Shutdown		(Accepted)	
In case of a power [failure], the [VIL] and all components of the outdoor system shall enter a safe state.					
Comments:		A power failure leads to a disruption of communication with the affected system. The higher-level communication partner must ensure that no unsafe state arises for all affected subsystems as a whole.			
Category:	SAR	Priority:	Mandatory		
Originator:	Customer	Origin:	==> [EL4 PH] [02_39] ==> [EL4 PH] [02_184]		
Target Release:	DIL_OBB_1.0.x DCA_OBB_3.1.0	Scope:	System		
Allocation:	DIL, DFE	Verification:	< will be added in a later document edition >		
		See also:			
Issues/Decisions:					
LH Text:		3.3.0-1 Bei Ausfall der Stromversorgung müssen alle betroffenen VSTWs mit allen zugeordneten Komponenten der Außenanlage einen sicheren Zustand einnehmen. 6.11.0-1 Bei Ausfall der Stromversorgung muss das VSTW und jegliche Komponenten der Außenanlage den sicheren Zustand einnehmen.			
PH Text:		Das Gesamtsystem DSTW wird nach den sicherungstechnischen Grundsätzen eines Fail-Safe Systems entwickelt. Der Ausfall der Stromversorgung führt zum Abbruch der Kommunikation mit dem betroffenen System, was beim jeweiligen Kommunikationspartner zur entsprechenden Sicherheitsreaktion führt. Bei Ausfall der Stromversorgung werden die VSTWs, sowie die Außenanlagenkomponenten (OCs und Umsetzer) analog zu einem Ausfall der Kommunikation des jeweils verbundenen Systems einen sicheren Zustand einnehmen.			

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EL4-DIL-SSS-REQ-00412 Safety - Data Configuration Process		(Accepted)
The supplier shall provide a certified [CENELEC]-compliant data configuration process for the [VIL] subsystem, the [OC] subsystems and all interfaces to the sub-systems and peripheral systems.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_188]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
	DCA_OBB_3.1.0	
<i>Allocation:</i>	DIL, VIL, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	6.13.0-1 Der AN muss einen zertifizierten CENELEC-konformen Datenkonfigurationsprozess für das System VSTW, Teilsysteme (OCs) und aller Schnittstellen zu den Teilsystemen und Umsystemen vorweisen.	
<i>PH Text:</i>	Das System VSTW und Teilsysteme (OCs) werden einen CENELEC-konformen Datenkonfigurationsprozess aufweisen, der insbesondere die Vorgaben der [OEVE/OENORM EN 50716] zur Entwicklung von Anwendungsdaten widerspiegelt.	

EL4-DIL-SSS-REQ-00413 Safety - Safety Case		(Accepted)
The validity of the safety case shall not be limited in time.		
<i>Comments:</i>		
<i>Category:</i>	SAR	<i>Priority:</i> Mandatory
<i>Originator:</i>	Customer	<i>Origin:</i> ==> [EL4 PH] [02_190]
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
	DCA_OBB_3.1.0	
<i>Allocation:</i>	DIL, VIL, DFE	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>	6.14.0-1 Die Gültigkeit des Sicherheitsnachweises darf zeitlich nicht beschränkt sein.	
<i>PH Text:</i>	Der Sicherheitsnachweis des Digitalen Stellwerks wird keine zeitliche Beschränkung aufweisen.	

3.1.7 MISCELLANEOUS

EL4-DIL-SSS-REQ-00654 RAMS - Diagnosis - Data for RAM Statistics		(Accepted)
The [DIL-CU] shall provide data from which RAM statistics can be created, based on the following events: → Starting of a [VIL Instance] → Stopping of a [VIL Instance] together with the reason for stopping → All state transitions of the remote control lines (separate for each line)		
<i>Comments:</i> • The [DIL-CU] does not need to create any such statistics itself. • This requirement is not derived from a customer requirement but taken over from other projects (e.g.[OMC]).		
<i>Category:</i>	MDR	<i>Priority:</i> Mandatory
<i>Originator:</i>	RAM	<i>Origin:</i>
<i>Target Release:</i>	DIL_OBB_1.0.x	<i>Scope:</i> System
<i>Allocation:</i>	DIL, VIL, ADM	<i>Verification:</i> < will be added in a later document edition >
		<i>See also:</i>
<i>Issues/Decisions:</i>		
<i>LH Text:</i>		
<i>PH Text:</i>		

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4. HISTORY

4.1 EDITION 01

4.1.1 PROPOSAL 01

This is the first version of the document.

This version covers the **RAM requirements** (according to [LH_PART_D] D2.2 “RAMS”) except for the chapters concerning the operational functionality.

Please consider the following review guidelines for the review of the document:

- The **allocation** of the requirements to subsystems and components will be reviewed separately. Feel free to make your comments already now.
- The **verification measures** of the requirements will also be reviewed separately. Feel also free to make your comments already now.
- **References and abbreviated terms** are marked with square brackets. They will be later replaced by hyperlinks (after the transfer to Polarion™).

The review **shall not result in a collection of hundreds of medium and grave comments**. If serious objections concerning the current structure and the content of the document exist, please let us clarify this better in an online discussion.

4.1.2 RELEASED

Version after Review.

A complete list of the differences can be found in the attached document “1257_Diff_Ed01Pr01_Ed01RL”.

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APPENDIX A DECISIONS

This chapter shall contain important decisions made in connection with the specified requirements.

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